LIFE Environment and Resource Efficiency

TECHNICAL APPLICATION FORMS

Part A – administrative information

LIFE 2015

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LIFE15 ENV/IT/000423

LIFE Environment and Resource Efficiency project application

Language of the proposal:

English (en)

Project title:

SMART WATER AND SOIL SALINITY MANAGEMENT IN AGRO-WETLANDS

Project acronym:

LIFE AGROWETLANDS II

The project will be implemented in the following Member State(s) and Region(s) or other countries:

Italy Emilia-Romagna
Spain Comunidad Valenciana
Cataluña

Expected start date: 01/09/2016 **Expected end date:** 30/06/2020

LIST OF BENEFICIARIES

Name of the **coordinating** beneficiary: Alma Mater Studiorum - University of Bologna

Name of the associated beneficiary: Agrisfera Soc. Coop. Agr. p.a.

Name of the associated beneficiary: Innovació i Recerca Industrial i Sostenible Name of the associated beneficiary: OSV SRL

LIST OF CO-FINANCERS

SECTOR

Resource Efficiency

LIFE15 ENV/IT/000423

TECHNICAL APPLICATION FORMS

Part B - technical summary and overall context of the project

SUMMARY DESCRIPTION OF THE PROJECT (Max. 3 pages; to be completed in English)

Project title:

SMART WATER AND SOIL SALINITY MANAGEMENT IN AGRO-WETLANDS

Project objectives:

Wetlands are highly productive, fragile ecosystems. This high productivity has led to the development of irrigated agriculture with increasing pressure on wetlands. Agri–wetland interactions must be precisely managed and precautionary measures must be taken in order to avoid that the soil use can significantly alter the soil functions. Specifically, soil salinization is a degradation process that is driven by irrigated agriculture in wetlands located in EU Med. regions, under arid and semi-arid conditions. Soil salinization negatively affects the agricultural productivity and alters the environment of the surrounding wetlands, whose natural value has been acknowledged by international treaties (Ramsar), EU Directives (Bird, Habitat) or by instruments of biodiversity conservation (European Ecological Network Natura 2000).

The pilot area proposed for the implementation of this system is located by the Italian N. Adriatic coast, in the Ravenna province, bordered by two systems of Natura 2000 areas, also included in the Ramsar list and belonging to the Regional Park of the Po Delta. The pilot area is affected by a remarkable soil salinization due to a combination of factors that make this area a good scenario to demonstrate the utility of the proposed technology.

Objectives:

- The main aim of this project is to develop an innovative system for water and soil salinity management in agricultural wetland ecosystems. This system will integrate a soil water and salinity wireless sensor network (WSN) with a decision support system (DSS) in a web environment. It will:

- save irrigation water, maintaining agricultural productivity, minimize soil and water quality degradation by salinization and allow the sustainable development of agriculture, particularly in the surroundings of valuable wetland habitats

- design and implement a WSN to control (i) amount and salinity of irrigation and drainage waters, (ii) soil salinity and water content and (iii) water-table depth;

- design and implement a web-based DSS that automatically scans the inputs from the sensor network to make irrigation recommendations;

- demonstrate the application of this technology for integrated water and soil salinity management in a Mediterranean agro-wetland, to ensure crop productivity and wetland conservation.

- contribute to the European Soil Thematic Strategy against soil degradation (COM (2006) 231, 22.9.2006)

- contribute to Habitat directive

- contribute to EUSAIR strategy (Pillar 3 Env. Quality)

Actions and means involved:

The actions are:

- B1: Environmental database construction, modelling of water and salt flow in the pilot area

- B2: Development of a new model of soil and water salinity for irrigation recommendation

- B3: Development and implementation of web-control system and user interface for irrigation management

- B4: Development of an integrated Wireless Sensor Network (WSN) module to monitor soil and water salinity, and water-table level for the sustainable irrigation

- B5: Field installation of wireless sensor network in pilot area

- B6: Field application of SMART-AGROWETLANDS II and evaluation of its effects on crops

- C1: Monitoring the Project´s impact on salinity status in agro-wetland pilot area

- C2: Monitoring the Project's impact on the natural and semi-natural habitats, plant community and biological quality of water and assessment on socio-economic impact

- D1: Communication Pack and Dissemination Activities

- D2: Networking and Training activities - Replicability

- E1: Project management, monitoring and After Life Plan

Action B1 will allow to delineate homogeneous areas of management within the pilot area and to know the corresponding soil and crop data. These data will be saved in a geo-database to provide input data for the SMART-AGROWETLANDS II software. Action B2 will allow to define a unifying criterion for characterizing both the water and salinity stress status of crops. This criterion will be the kernel of the SMART-AGROWETLANDS II software which will be developed next in the same action. Action B3 will allow to have the web-control system validated and able to scan the inputs from the sensors, and also to run the SMART-AGROWETLANDS II model. The elements for communication in the WSN will be developed in Action B4. The communications will be checked which will be necessary to the subsequent deployment of sensors in Action B5. As a result of this action the DSS will be ready to work, which will occur in Action B6.

The system performance will be checked in Action C1. The project's impact on the natural and seminatural ecosystems will be checked during Action C2. Communication and dissemination for projects results will be the objects of actions D1.

Networking and training activities aiming at replicability and transfer activities will be the object of action D2. Action E1 will focus on management, reporting and quality assurance.

Expected results (outputs and quantified achievements):

R1

The main achievement of the project will be the development of a DSS for sustainable water and soil salinity management, allowing farmers and water managers to adopt the most adequate irrigation plans, fulfilling agricultural and ecological requirements. In the framework of the climate change, the water management optimization will improve crop production, mitigate soil salinity/salinitazion and maintain biodiversity of wetland habitats.

The crops growing in the pilot area tolerate soil salinity ranging from 3.4 dS/m for alfalfa (most sensitive) to 9 dS/m for barley (most tolerant). A significant expected result will be to keep soil salinity below the harmful level for crops of medium tolerance to salt, taking in consideration the crop species with a wider presence and more suitable for the cultivation in the pilot area itself.

R2

As a consequence, a better production for the species with medium or high tolerance to salt is expected, and, at the same time, a reduction of the amount of water use for crop production. Specifically, soil salinity and water use are expected to decrease by 20% and 15% respectively in the fields and the drainage water flow, and salinity are expected to be kept within ranges adequate for the conservation of the wetland habitats, sensitive to high soil and water salinity. Measurements of these outputs will be made to check the results of the application of the SMART-AGROWETLANDS II system.

Other intermediate expected results are:

R3

Integration of the on-line measurements from various sensors: i) soil water content; ii) soil salinity; iii) irrigation and drainage water salinity; iv) water-table depth; v) soil properties, within a single DSS.

R4

Optimization of the number and allocation of soil sensors for monitoring soil salinity within a large agricultural area, by means of geostatistics techniques and numerical modeling of water flow and salt mobilization. A cost reduction of at least 30% in sensor acquisition and implementation is expected.

R5

Up-scaling of water management from plots to irrigation sectors, considering the water quality and the impact on the surrounding environment, reducing water footprint by 20%.

Is your project significantly climate-related? Yes [X] No [ ]

Water, soil, soil salinity, agriculture, wetland areas, are significant keywords for this project, strictly related with climate and the foreseen climate change.

Climate change will have widely differing effects on the European water resources. Average run-off in Southern European rivers is predicted to decrease. Some river basins that already face water constraints, may see marked decrease of water availability. Low water and droughts will have severe consequences, particularly on agriculture, irrigated agriculture, forestry, energy, and drinking water provision. Water management is also expected to play an increasingly central role in adaptation to climate change as indicated in the European Climate Adaptation Platform (www.climateadapt.eea.europa.eu/home), where agriculture is considered a vulnerable key sector.

Soil is another important resource influenced by climate. Healthy soils can mitigate climate change, but not all soils are equal in their capacity of carbon storage. Soils in warm and dry areas of Southern Europe contain less carbon than grassland soils or peatland soils of Northern Europe. Climate change is believed to result in an overall deterioration of soil quality in Mediterranean regions. Possible effects include salinization, organic matter loss, increased wind and water erosion. Primary salinization caused by climate change (increases in temperature and decrease in precipitations) can be increased by human and agricultural practices, that can induce a salinization of secondary origin. Inappropriate irrigation, with salt-rich irrigation water, because of the scarcity of water of good quality, is one of the main causes of secondary soil salinization in Mediterranean area, where the problem of soil salinization is getting worse. The pilot area in Italy and other Mediterranean areas in Spain, where networking and replicability activities will be performed, fall within these critical zones.

Wetland systems are vulnerable and particularly susceptible to changes in quantity and quality of their water supply, and it is expected that climate change have its most pronounced effect on wetlands, through alterations in their hydrological regimes. Future sea level rise, increased storm severity, subsidence of coastal areas, rise of the water-table, saltwater intrusion into coastal aquifers are expected to have a significant impact on the wetlands. The sectors that depend on the good and services they provide, will also be threatened.

Is your project significantly biodiversity-related? Yes [X] No [ ]

Water, soil, soil salinity, agriculture, wetland areas, significant keywords for this project, are strictly related with biodiversity.

The project is related to biodiversity from a general point of view. It concerns in fact an efficient use of not renewable environmental resources, such as water and soil, prevents the degradation of their quality, promotes a sustainable agriculture and explores new possibilities for the precision farming.

More specifically, the project operates in a context where agriculture and protected areas of the ecological Network Nature 2000 (CSI/SPA IT 4070005 Pineta di Casalborsetti, Pineta Staggioni, Duna di Porto Corsini) are close each other, and where real threats to the survival of the variety of wetland habitats, due to soil salinization, are present.

The project recognizes the close link between the agricultural areas, obtained from reclaimed wetlands, and the natural and seminatural surrounding wetlands, not used for agricultural scopes. It gives an unifying reply to agricultural requirements and biodiversity conservation requirements. The approach adopted is in line with the strategies of maintaining biodiversity, undertaken by the European Community, through the primary tool of ecological networks (Habitat Directive 92/43/EEC and the Natura 2000 Network). Ecological networks integrate anthropic spaces (agricultural areas) with natural and semi-natural areas, reconciling the needs of development, with those of maintaining the natural heritage. The sustainable agriculture proposed by the project allows the maintenance of biodiversity in the surrounding natural a semi-natural environment.

Through water flow and salinity maintained within the ranges needed to ensure the conservation of habitats in the wetlands (see “expected results”), many of the threats on the wetland habitats will be mitigated or eliminated, helping the biodiversity of habitats and species.

The study area is therefore a very interesting pilot area, where the compatibility between a sustainable agriculture and biodiversity conservation will be experimented.

The proposal addresses the following project topic(s):

- Projects designed to achieve better soil management (decreasing erosion, maintaining soil organic matter, avoiding compaction and contamination, conserving/restoring carbon rich soil, etc.) at the local, regional or national level. The methods used can include monitoring tools and practices or the improvement of administrative and legal frameworks. Of particular interest will be projects providing cost-effective solutions improving existing tools or methodologies, or supporting soil functions as part of the wider ecosystem, e.g. for water retention.

Reasons why the proposal falls under the selected project topic(s):

An efficient use of the water and soil, through a smart management of these natural resources, is the main topic of the SMART-AGROWETLANDS II project. Farmers and water managers will be able to adopt the most adequate irrigation plans in agro-wetlands, through a DSS developed during the project. Water management optimization, experimented in Mediterranean agrowetlands, should improve crop production, mitigate soil salinity/salinization and maintain biodiversity in natural and semi-natural surrounding wetlands. An efficient use of water and soil resources allows to fulfill agricultural and ecological requirements. An efficient use of water and soil resources will be in turn achieved through the adoption of the Precision Agriculture (PA) approach, which improves the agricultural process as well as the protection of natural resources. PA “applying the right treatment in the right place at the right time” increases an efficient use of the resources and the maintenance of natural capital, the core elements of the EU 2020 growth strategy (http://ec.europa.eu/europe2020/index_en.htm)

The project will produce a cost effective on-line monitoring tool for: irrigation water volume and salinity, pore water and soil salinity, water-table depth, drainage water volume and salinity, water requirements of crops, soil hydraulic properties, leading to a better soil and water management.

The project will be implemented at the regional scale although the developed systems and tools will be exportable to the national and European level, allowing a smarter, more accurate and improved soil and water management.

The project, in its first elaboration, on the same topic, **was already proposed and approved as LIFE13 ENV/ES/001125** (LIFE SMART-AGROWETLAND – SMART WATER AND SOIL SALINITY MANAGEMENT IN AGRO-WETLANDS), to be tested in the Carrizales pilot area (Helche, Spain). However, because of management difficulties of the former project coordinator, the **Grant agreement wasn't signed and the project cancelled without starting**, and the test wasn't done and wasn't financed for the partners' and lead partner's decision. After two years the partnership has been recreated under a stronger lead partner able to guarantee the necessary continuity and management, a new and more appropriate pilot area has been identified and several improvement have been implemented thanks of the two years preparation time.

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ENVIRONMENTAL PROBLEM TARGETED

Wetlands are amongst the most environmentally sensitive areas in the European Union territory. Although nowadays wetlands amount to less than one per cent of the total EU area, they are among the most productive and fragile ecosystems. In the World, over 50% of wetlands were lost in the past century, and the remaining wetlands have been degraded due to human activities (Fraser and Keddy, 2005) . Several factors contribute to the vulnerability of wetlands in Mediterranean regions. First, they are generally located in low areas, where the drainage outlet is restricted. Because of such topography, salts are concentrated in the underlying phreatic waters leading to salinization of soils and waters. Second, the arid to semiarid climate with high evapotranspiration and low precipitation rates, that domains Mediterranean regions, increases the evapo-concentration of salts. An increase in water and soil salinity on coastal wetland areas has direct effects on agriculture (FAO, 2002; Wallender and Tanji, 2011 ; www.fao.org/DOCReP/003/T0234e/T0234E01-html) as well as on the natural habitats (Antonellini and Mollema, 2010; Greggio et al., 2012; Bouzourra et al., 2015). These effects include, among others, the decrease in plant species richness, decrease in wetland dry biomass, changes in plant communities, habitat loss for plant and animal species and increase in their mortality.

Since wetlands in European Mediterranean regions are highly productive, they were historically exploited for hunting, which protected these areas from further degradation. However, the high carrying capacity of wetlands also prompted an early colonization by farmers. The agro-wetlands so created cannot completely replace the functions of natural ones (Desrochers et al., 2008), however, it is widely recognized that artificial wetlands can provide alternative habitats for flora and fauna (Connor and Gabor, 2006).

In other hand, salinization is one of the main soil degradation processes, as indicated in a European directive (COM 232/2006). This process is highly important in soils under irrigated agriculture and semiarid climate, which are frequent in Mediterranean agro-wetlands ecosystems.

The agro-wetlands require a special and detailed regulation of water flow and water quality monitoring to ensure the conservation of the wetland habitats and to sustain the agriculture activities. The main factors that should be monitored and controlled, to know wetland status are i) water-table depth, ii) water level fluctuation, iii) vegetation, iv) salinity, v) topography, vi) food type, vii) food accessibility, viii) wetland size, and ix) wetland connectivity.

Specifically, salinity affects agro-wetlands mainly in two different ways:

- To soil. The agricultural yields decrease as a consequence of the stress on the plants by the soil salts. To mitigate this stress, higher amounts of water are required to sustain the production and reduce the soil salt build-up. Under arid and semiarid climates, crop water requirements are higher, waters used for irrigation are usually more saline than in more humid areas, and thus, an extra amount of water is required to wash the salts out of the soils. In these areas soil salinization becomes one of the main soil degradation processes exacerbated by agricultural activities and, eventually, leading to desertification.

- To wetland. As indicated by Ma et al. (2010), salinity is one of the main factors that regulate the wetlands. This factor is interlinked with others as water level fluctuation, water depth and vegetation. The water management in the agro-wetlands affects the water level and salinity. Usual irrigation scheduling is aimed at attaining higher water-use efficiencies at field scale. As a consequence of the water savings, lower volumes of drainage water with higher salinity are produced. In wetlands whose main feeding waters are from agricultural origin, this practice decreases the water level, increases the water salinity and, consequently, leads to losses of flora and fauna biodiversity and their replacement by few species more salt tolerant.

In agro-wetlands in case the soil salinity in the surrounding irrigation fields increase up to a level beyond which cropping is not profitable any more, agricultural activities could be abandoned. This fact would reduce water flow to wetlands, which, in turn, would lead to clogging of drainage channels by sediments and vegetation. As a consequence the habitat for the fauna could be dramatically affected. Due to the reduction in water flow, the permanent pools and marshes could turn into temporary lagoons, and consequently an important number of species of birds, fishes and flora would face dramatically reductions of their populations by the scarcity and high salinity of the water. Birds that drink saline water lose body weight by dehydration (Hannam et al., 2003). Besides, the halophytic plant communities of salinized ecosystems provide less adequate sites for bird roosting and nesting.

Overall, water management in wetlands requires a holistic view of the wetland agro-ecosystem. The tight interrelationship between agriculture and the wetland through the water and salt flows, as shown

in Figure 1 demands an integrated control that takes into account the agricultural water flows, and soil, and water salinity. Such integrated control is necessary to improve both the habitat quality of the wetland ecosystem and the agricultural profitability.

In this context, a control system designed to optimize the water management for the simultaneously fulfilling of the crop water requirements, the mitigation of soil salinization and the conservation of the wetland habitats is required.

The pilot area, where this system will be designed and implemented, is located in the province of Ravenna (Italy), between the Reno river to the North and the Lamone river to the South and close to the Northern coastline of the Adriatic Sea (Figure 2). In particular, the field activities of this project will be performed in five different farms of the co-operative Agrisfera (Figure 3): Baronia (48 ha), Marcabò East (183 ha), Marcabò West (220 ha), Augusta (90 ha) and Chiavichino (68 ha), for a total of 609 ha. Soil salinization affects all the five farms.

In the territory surrounding the pilot site, besides the agricultural and inhabited areas, eight CSI/SPAs of the European Ecological Network Natura 2000, belonging also to the Regional Park of the Po Delta, are present. They host a variety of terrestrial and aquatic habitats, these latter with salty, brackish and fresh waters. Some of these habitats are partly compromised by the changes taking place on the water quality and by the soil salinization.

The eight protected areas are shown in Figure 4. On the whole they form two systems oriented NorthSouth, reciprocally parallel and parallel to the coast, that delimit at West and East side a wide agricultural strip of quite recent land reclamation, where the five farms of the co-operative Agrisfera are placed. The activities of the C2 Action of this project will be performed in some of the eight protected areas.

The agricultural areas of the pilot site, and the surrounding ones, are mostly irrigated and cultivated with winter cycle cereals (wheat, barley, oat), springer-summer cereals (corn, sorghum), as well as with rapeseed, sunflower, alfalfa and vegetables, by some hundreds of farmers organized in co-operatives. Crops such as barley and wheat are tolerant or quite tolerant of the soil salinity, sorghum and corn are more sensitive, whereas alfalfa is the most sensitive.

The quality of water in the irrigation channels is quite poor during the winter, for the considerable salt content (Cl-112÷988 mg/l; SO4--70÷120 mg/l; Na+ 54÷75 mg/l), while during the spring-summer period improves (Cl- 27÷79 mg/l; SO4-- 21÷201 mg/l; Na+19÷29 mg/l), as result of the intake, through CER (Canale Emiliano-Romagnolo), of good quality water coming from the Po river. However, not all agricultural areas of the district, such as those closer to the coast, profit by the summer improvement of water quality, because they are not irrigated. In these cases the soil salinity may reach very high values, so as to exclude any summer crop. In these conditions the measured electrical conductivity of soil saturation extract is always higher than 2.2 dS/m and can reach 4.8 dS/m at the top soil layer.

The soils in this area have been formed on sandy dunes and on finest materials (e.g. silt and clay) that are derived from alluvial deposition of the both rivers Lamone and Reno. The soil map 1:50k of the Emilia-Romagna Region (Figure 5) recognizes four types of soil complex: Marcabò soil – MCB1; Savio soil – SVO1; Galisano soil – GLS3; Cerba soil – CER1. The water-table is found at 65 ÷ 70 cm depth, developing hydromorphic and aquic soils. These conditions of high salinity water, silt-clayey soil texture, shallow piezometric level and semiarid climate, lead to an important risk of soil salinization (Figure 6).

In the Ravenna province, very active naturalistic and cultural associations are involved in the environment knowledge and conservation and cultural heritage of the territory:

Società per gli Studi Naturalistici della Romagna, www.ssnr.it/ita.htm;

Gruppo Micologico Ravenna, www.micologia.racine.ra.it;

ARCA, www.larcapuntealberete.it;

Associazione Culturale Erbe Palustri Bioregione Valle Lamone, www.erbepalustri.it .

These associations organize periodical meetings, are involved in international educative projects (SiiT Project, for the Cooperation Italy-Slovenia) or in the valorization of the economic utilization of the

wetland areas. This indicates the high social concern of the local populations to conserve the agrowetlands of their territory, and their natural and cultural heritage.

PROJECT PILOT / DEMONSTRATION CHARACTER OF THE PROJECT

The project, in its first elaboration, on the same topic, was already proposed and approved as LIFE13 ENV/ES/001125 (LIFE SMART-AGROWETLAND – SMART WATER AND SOIL SALINITY MANAGEMENT IN AGRO-WETLANDS), to be tested in the Carrizales pilot area (Helche, Spain). However, because of management difficulties of the former project coordinator (change of the management board), the Grant agreement wasn't signed and the project cancelled without starting. So, the test wasn't done and wasn't financed for the lead partner's decision. After two years the partnership has been recreated under a stronger lead partner able to guarantee the necessary continuity and management, a new and more appropriate pilot area has been identified, while several improvement have been implemented thanks of the two years preparation time. For this reason this project is not a clone, but the same project, improved and submitted again by a stronger partnership, in continuity with the previous shape.

The main innovative aspect of this project is still the integrated control of water (amount and quality) and soil (water content and salinity) in agro-wetlands. Generally, water management systems in agriculture are focused on the supply of enough water to the crops. Innovative irrigation increasingly makes use of soil water control systems by sensors to perform this task. Soil salinity, however, has not been fully integrated in modern irrigation management. Only general ideas or, at best, simulation models based on the irrigation water salinity are used. Up to date, soil salinity monitoring by sensors, and recommendation systems based on their data have not been developed yet.

Since natural wetlands are usually tightly linked to agro-wetlands in the European Mediterranean regions, the water salinity and the outflow water amounts from the agriculture fields must be integrated within a single water management system. In this sense this project is also innovative.

This project builds on the results of the very successful FP7-SME-2011-WATERBEE-DA (Grant Agreement 283638) and SMARTPADDY (Grant Agreement 285861) projects which have very convincingly researched, developed and proved the concept of low cost Wireless Sensor Networks for on-line remote measurement of salinity in paddy fields and for irrigation management in agriculture. Both novel technologies are now ready for demonstration to the agricultural communities.

In the WATERBEE project, a wireless sensor network in combination with a web decision support system was developed to optimize the irrigation management in agricultural fields. The sensors have been installed in several points within the irrigation district to measure soil water content at different soil depths, and send the information to a central system where irrigation recommendations are provided to the farmer or centrally to the irrigation engineer. This system has been installed, and is in use at 14 reference sites across Europe in Estonia, Malta, Italy, Spain, Sweden and the United Kingdom.

In the SMART-PADDY project, which ended in September 2013, the main objectives aimed at the development of a simple wireless low-cost sensor network to accurately measure in real time the water salinity in irrigation and drainage channels of rice-paddies. The consortium of this project is composed by 9 project partners from 4 EU Member States (France, Italy, Spain, Greece) plus Turkey. The system has been installed in four different rice-growing areas across the EU (Camargue, Delta Guadalquivir, Delta d’Ebre and Thessaloniki) where proximity to the sea, and salinity in irrigation waters cause low crop quality and yield losses.

IRIS and OSV, both members of the previous projects, will participate in this LIFE proposal as associated beneficiaries in order to provide the full technological assistance to this demonstration action.

The UNIBO participation in the SMART-AGROWETLANDS II improves the project pilot/demonstration character, because of the consolidated and updated experience of UNIBO on the different knowledge required by the project. In the framework of this LIFE project, the experience of the DipSA research groups, integrated with the experience of the DICAM groups is an innovative element that reinforces the planning of the field work, the experimental investigations and the modeling.

The Department of Agricultural Sciences (DipSA) of UNIBO has a wide and articulated experience on the functioning of agricultural systems, as well as on natural and semi-natural plant communities. The Soil Science research group is from many years involved in integrated and innovative studies concerning the physicochemical and biochemical interrelationships within the water-soil system. Different aspects on the ecological functionality of soil processes, such as the activity of soil microbial

biomass and the carbon sequestration, as well as the soil degradation due to loss of biodiversity have been investigated. The group has wide and depth knowledge on various aspect concerning the salinity and pollution in soil and waters of the Ravenna coastal plain (Buscaroli et al., 2009; Petrini et al., 2014; Canfora et al., 2015; Sanoubar et al., 2015). The Field-Crop research group has a relevant experience on the assessment of field-crop growth in stressed and un-stressed conditions, as well as on the evaluation of crop performances (Gabellini et al., 2006; Barbanti et al., 2007; Ahmad Sher et al., 2013; Cattaneo et al., 2014).

The Department of Civil, Chemical, Environmental and Materials Engineering (DICAM) of UNIBO has a wide and articulated experience on Hydraulic and Irrigation Engineering as well as on Geomatic, Photogrammetry and Remote Sensing.

The hydraulic group was involved in several Research Projects funded by Local Authorities on topics for the management of water and diffused pollution in rural areas. In particular the researchers involved in this proposal had worked in the project RIMEDIA (2007-2008) funded by Po River Authority (Parma, Italy) finalized to the monitoring and modeling of the irrigation and drainage artificial channels of the Reggio Emilia Province both for hydraulic and water quality aspects. The monitoring activity was carried out by automatic 24 bottles samplers and several water level gauges. In this Project, in collaboration with the Italian National Research Council (CNR), the effect of narrow buffer strips in nitrogen removal was also evaluated (Balestrini et al., 2011). Another Project, funded by EmiliaRomagna Region (Italy) in 2007, with title: "Sediment characterization of artificial channels in the territory of the Province of Bologna", was addressed to the monitoring and modeling of sediment and water quality aspects in irrigation channels (Ying et al., 2012; Liserra et al., 2013; Gaeta and Lamberti,
2015) . Some years before, the group was involved in two wide scale projects funded by MAF (Agriculture and Forestry Ministry): “Po AcquAgricolturAmbiente” (Po Water Agriculture and Environment) on water management for agricultural purposes in the Po plain, and circulation of pesticides and nutrients in agricultural soils, performing analyses and modeling of water quality, both carried out by the CER Consortium, managing the largest irrigation system in the Po plain and Emilia Romagna region and an Agriculture research service aimed at improving irrigation efficiency.

The Geomatic group (Bitelli et al., 2011; Vittuari et al., 2013; Bitelli et. al., 2015) is involved in the realization of remote systems for kite/balloon-based high resolution photogrammetric surveys, the use of satellite remote sensing, photogrammetry and digital cartography, in the study of territorial changes as well as on change detection applications - not only related to a qualitative evaluation, but also to quantitative data extraction. The group participated in EU funded projects: "SubCoast - A collaborative project aimed at developing a GMES-service for monitoring and forecasting subsidence hazards in coastal lowland areas around Europe " (FP7-SPACE-2009-1, 2010-2013); "EnergyCity Reducing energy consumption and CO2 emissions in cities across Central Europe" (EU Central Europe, 2010-2013), WP leader. The group is now leading “ChoT - *The challenge of remote sensing thermography as indicator of energy efficiency of buildings”,* a MIUR financed SIR 2014 project.

All these experiences will be useful to the proposal, both for monitoring and modeling activities

Demonstration character:

One of the derived objectives of this project is to demonstrate to farmers, irrigation managers, agricultural co-operative staff, conservationists, policy makers and other stakeholders how a technology based on the integrated use of sensors along with a web-based irrigation recommendation system can be useful to simultaneously i) reduce the amount of water used for irrigation in the crop production, ii) to protect soils against salinization processes, and iii) to avoid habitat degradation of the associated wetlands by high salinity in the drainage waters.

The relative small extension of the estate of the co-operative farm Agrisfera (4000 ha), of which only 600 ha were selected for the field experimental work of this LIFE project, can be considered as a Mediterranean wetland area at a pilot scale. The aim of working in a small area is to develop a wireless sensor network (WSN) and a web-based decision support system that can be handled by just one person. In spite of this small size, the selected area reflects all the agro-natural wetland interactions that usually are found in Mediterranean environments, in terms of salinity, climate, soil, crop production and natural habitat. Therefore the actions envisaged in the project will be conducted at this pilot scale, in order to make them extensive to other similar agro-wetlands throughout Mediterranean Europe in future actions or projects different from this.

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Figure 2 - Localization map of the pilot area, see above, and Figure 3 - Five different farms of the co-

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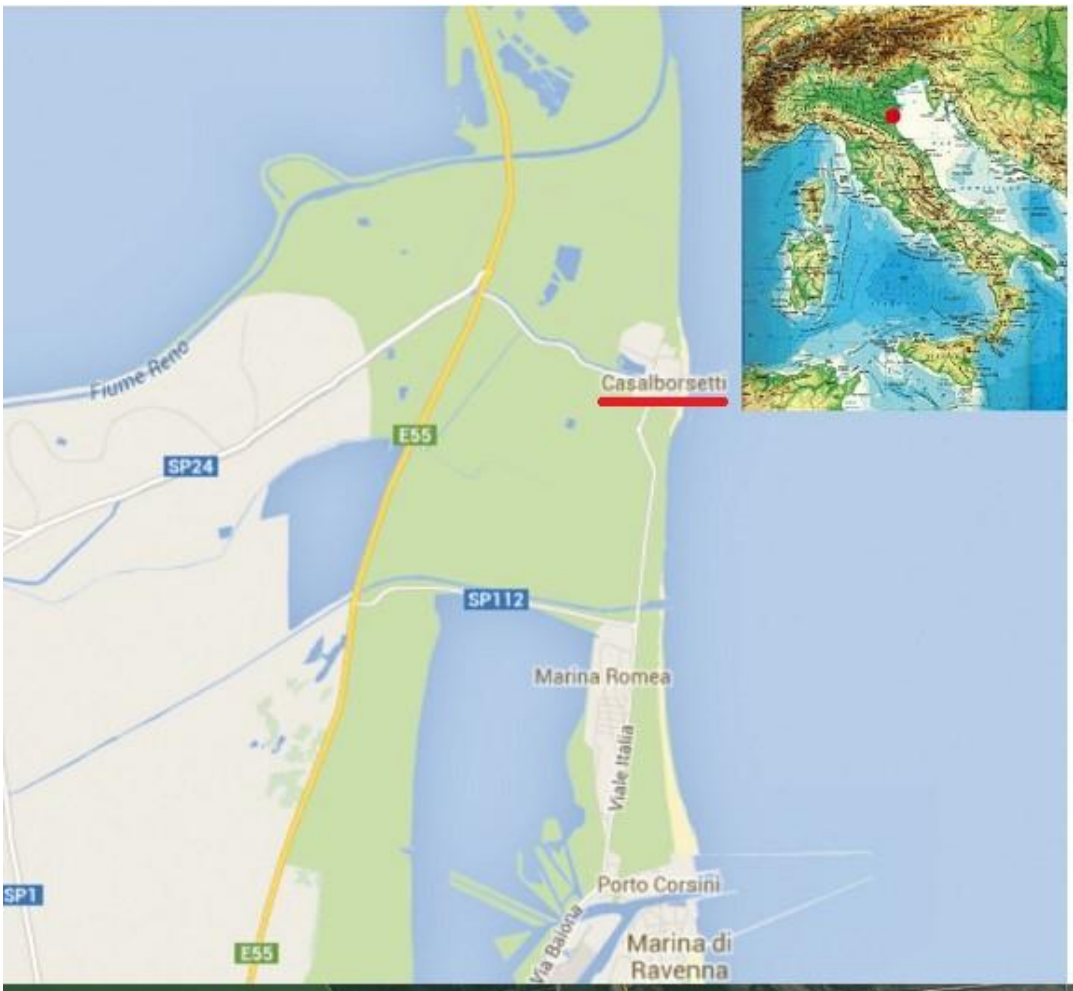

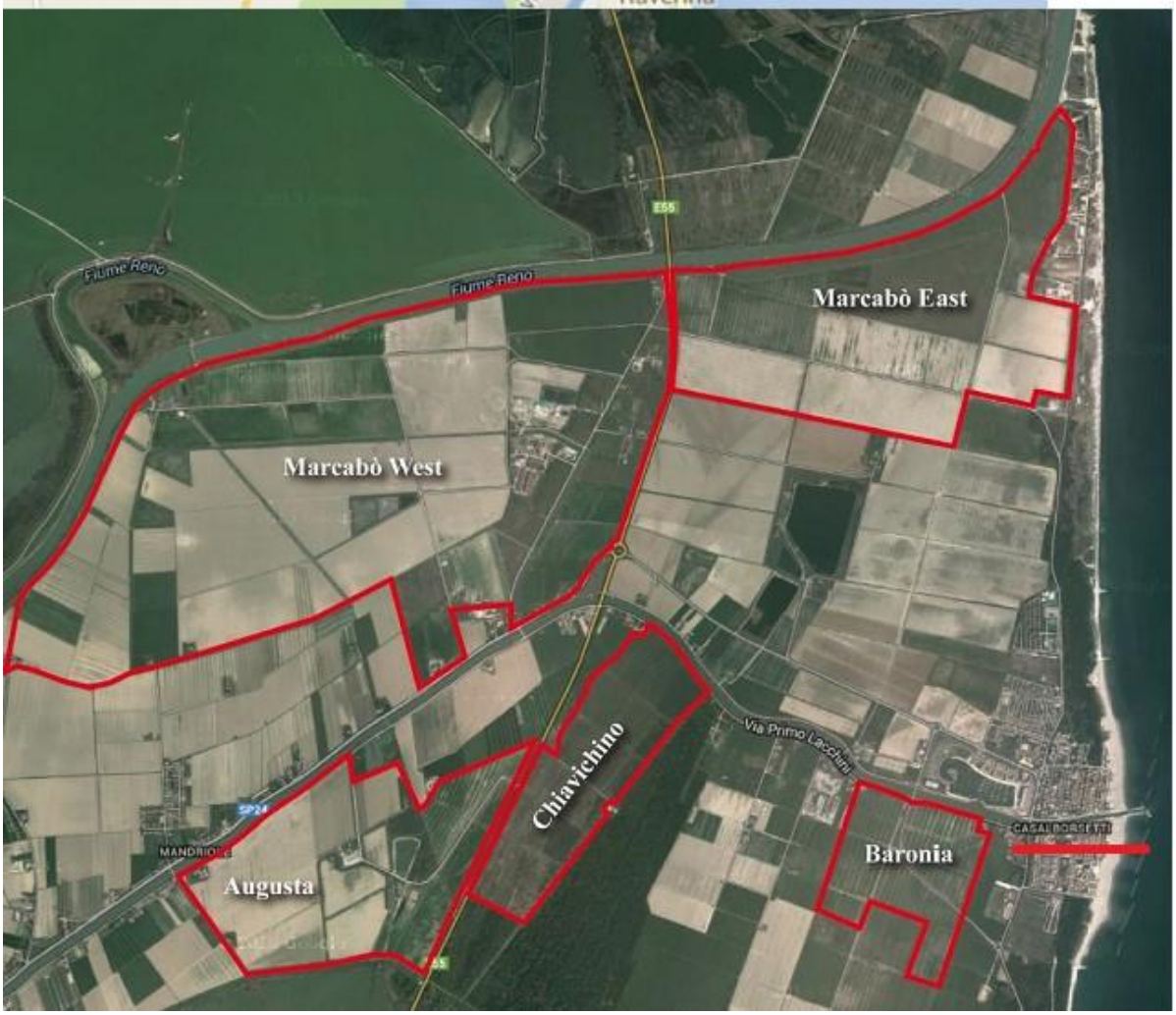

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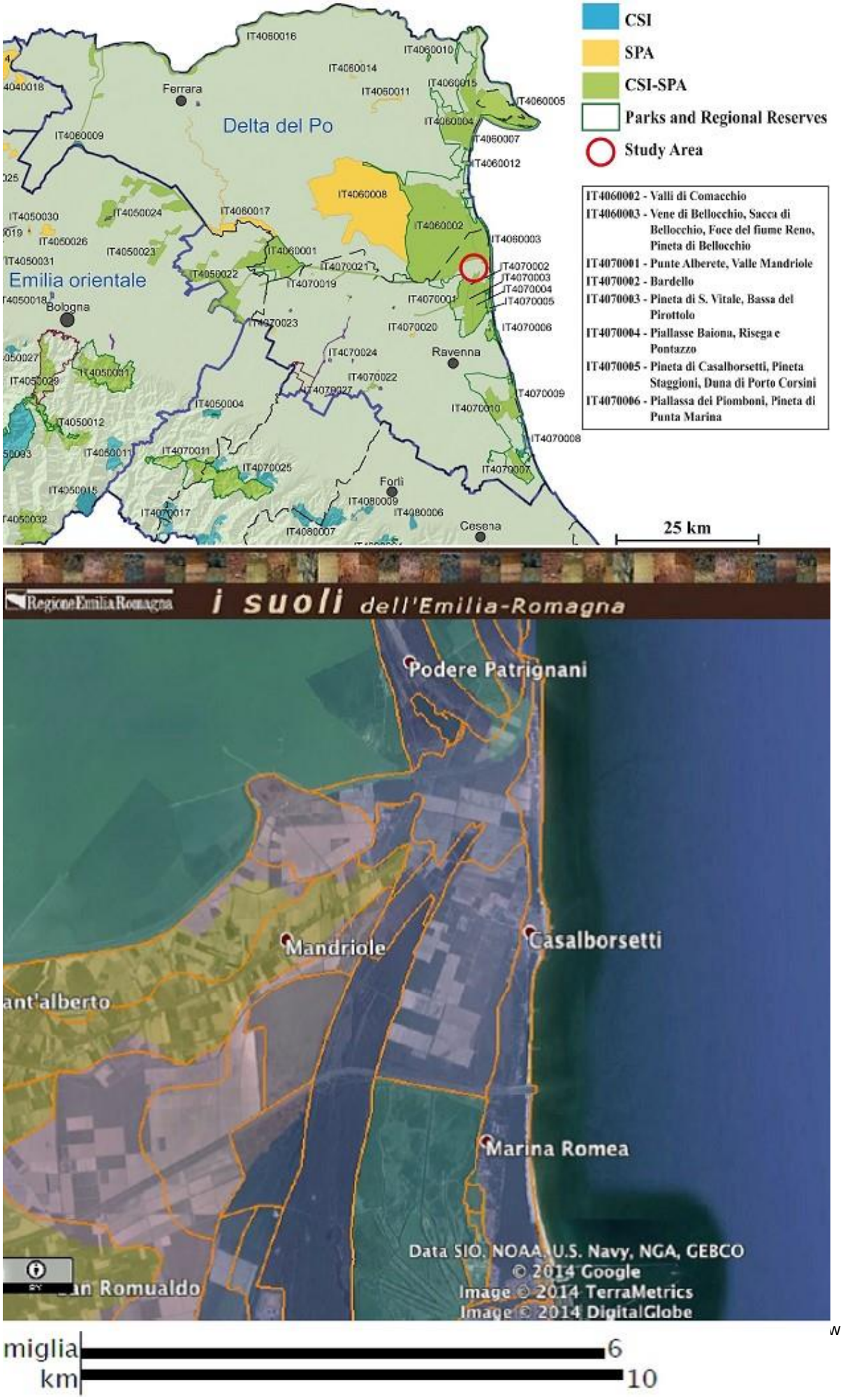

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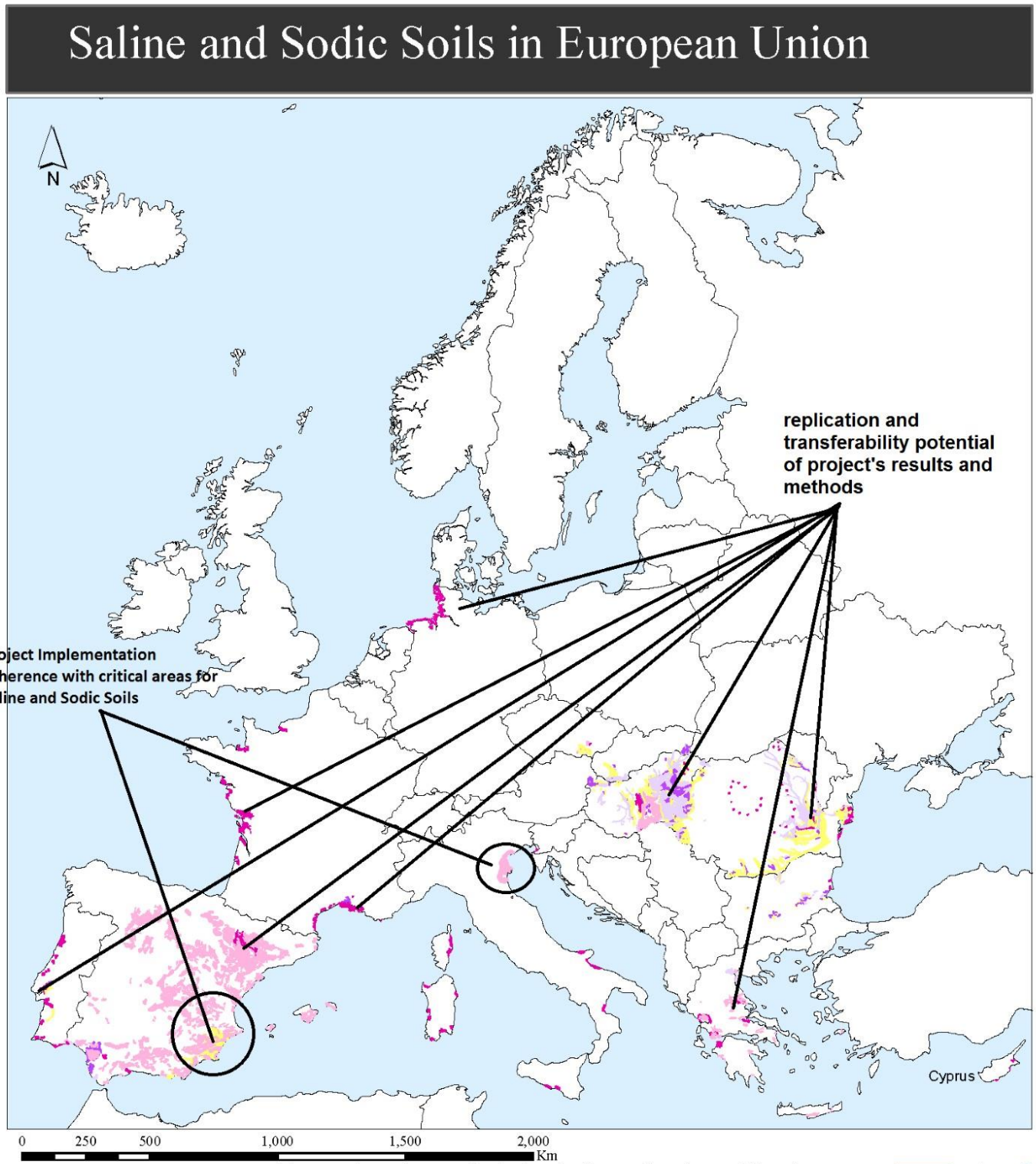

Saline and Sodic Soils

Legend

- Saline > 50% of the area
- Sodic > 50% of the area
- Saline < 50% of the area
- Sodic < 50% of the area
- Potentially salt affected area

The map shows the area distribution of saline, sodic and potentially salt affected areas within the European Union. The accuracy of input data only allows the designation of salt affected areas with a limited level of reliability (e.g. < 50 or > 50% of the area); therefore the results represented in the map should only be used for orientating purposes.

MAP INFORMATION

Spatial coverage: 27 Member States of the European Union where data available.

Pixel size: 1km
Projection: ETRS89 Lambert Azimuthal Equal Area

Input data - source
Soil data - European Soil Database v2
1:1.000.000 scale Map of Salt Affected Soils in Europe (Szabolcs) 1974)

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Digital datasets can be downloaded from http://eusoils.jrc.ec.europa.eu/

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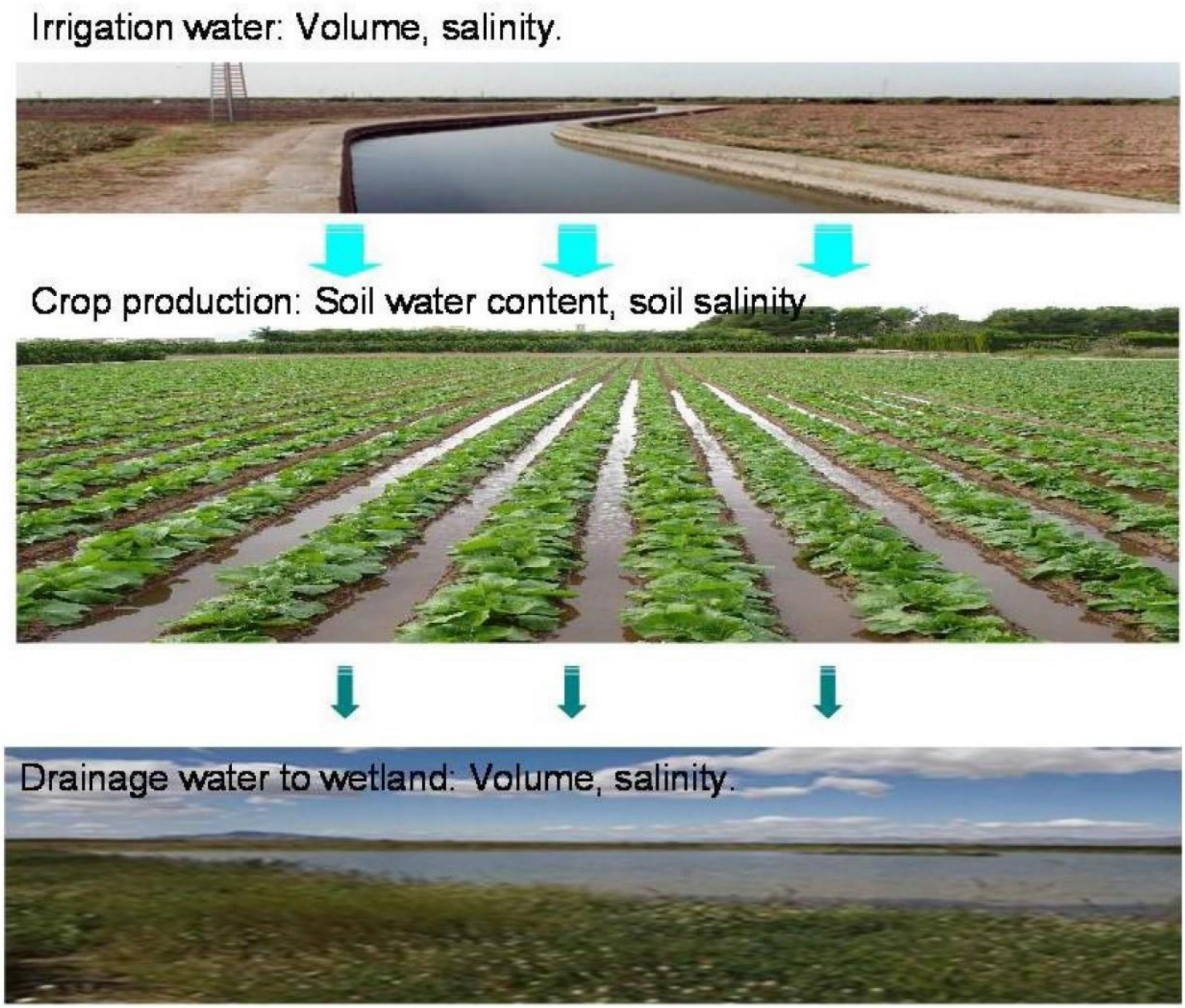


ecosystem

EU ADDED VALUE OF THE PROJECT AND ITS ACTIONS

The project will develop a DSS for sustainable water and soil salinity management, allowing farmers and water managers to adopt the most adequate irrigation plans, fulfilling agricultural and ecological requirements. In the framework of the climate change, the water management optimization will improve crop production, mitigate soil salinity/salinization and maintain biodiversity of wetland habitats.

This proposed LIFE action adds value to the results obtained in two European research projects (WATERBEE and SMART-PADDY) and goes beyond. It additionally provides environmental technological solutions to mitigate the soil salinization, which has been considered in the European Soil Thematic Strategy against soil degradation (COM (2006) 231, 22.9.2006) as one of the methods adequate to ensure the sustainable use of soil. Besides, the European Commission indicated that the soil salinization will be exacerbated by climate change.

Additionally in agro-wetland ecosystems, not only the agricultural part, but also the habitat conservation for flora and fauna has to be taken into account. The Habitat European Directive-HD (92/43 /EEC) enacted in 1992, provides the framework to protect the flora and fauna as well as their natural and semi-natural habitats. The Nature 2000 ecological network was created as a consequence of the application of this Directive.

Nevertheless, the Habitat Directive was overtaken in 2000 by the Water Framework Directive-WFD (2000/60 /EC) establishing a framework for European Community action in the field of water policy. In the WFD environmental quality is understood in a holistic way and therefore, it includes not only all aspects of water quality (biological, hydro-morphological and physical-chemical), but also the relationships amongst them and with the rest of the environment. As a consequence, the WFD considers aspects collected on previous directives, not strictly focused on water protection such as the Habitat Directive, builds on them, and goes beyond. Specifically, in Article 6.1, the WFD stipulates the establishment of a register of protected areas "which have been designated as requiring special protection under specific EC legislation for the protection of their surface water and groundwater or for the conservation of habitats and species directly depending on water". The register must contain i.e. "areas designated for the protection of habitats or species where the maintenance or improvement of the status of water is an important factor in their protection, including relevant Natura 2000 sites […]" (Annex IV, (v) WFD).

Any Natura 2000 site with water-dependent (ground and/or surface water) included in Annex I habitat types or Annex II species under the Habitat Directive or with water-dependent bird species of Annex I or migratory bird species of the Bird Directive, and, where the presence of these species or habitats has been the reason for the designation of that protected area, has to be considered for the register of protected areas under WFD Art. 6.5. These areas are summarized as "water-dependent Natura 2000 sites".

Since the objectives of this project are focused on the soil quality, water use efficiency and the wetland habitat conservation, they are tightly associated to the first two aims of the WFD which are: (a) “to prevent further deterioration and to protect and enhance the status of aquatic ecosystems and, with regard to their water needs, terrestrial ecosystems and wetlands directly depending on the aquatic ecosystems”, and (b) “to promote the sustainable water use based on a long-term protection of available water resources”. The development of the SMART-AGROWETLAND management system will address these two main aims of the WFD, i.e., to increase the water use efficiency for sustainable use (save water), while simultaneously reducing the soil salinity and protecting the wetland habitat and the aquatic ecosystem.

The WFD is the most substantial piece of EU water legislation thus far and aims to overcome the fragmentation of European water policy.

In this context, framed by three different European directives (habitat, soil and water), is where this LIFE project proposal arises: the development of a technology for the integrated management of soil and water within an agro-wetland environment.

The development of the wireless web-based system of this proposal can be easily transferable to other European agro-wetland areas where the current agricultural practices, climate conditions, and restricted soil drainage pose a high soil and water salinization risk.

There are many similar wetlands affected by agricultural activities across European Mediterranean countries (see Figure 7), which could benefit from this technology. Among these:

- Italy: “Daccia-Botrona”, “Bolgheri” and “Lagoon of Orbetello” in Tuscany, “Saline di Margherita di Savoia” and “Torre Guaceto” in Apulia, “Stagno di Cagliari”, “Stagno di Cabras” in Sardinia…

- France: The Rhône River Delta (Camargue)…

- Greece: Amvrakikos gulf, Axios, Loudias and Aliakman River Delta, Evros River Delta, Koychi lagoons, Lake Vistonis…

- Spain: Ebro River Delta, Albufera de Valencia, Almenara Marsh, Pego-Oliva Marsh, Pantano del Hondo de Elche, Lagunas Torrevieja, Lagunas Santa Pola, Prat de Cabanes-Torreblanca, Aiguamolls de l'Empordà Natural Park, Marismas Guadalquivir etc… We have once more to remember that the actions to be tested in Spain in Carrizales area through the project LIFE13 ENV/ES/001125 (LIFE SMARTAGROWETLAND – SMART WATER AND SOIL SALINITY MANAGEMENT IN AGRO-WETLANDS) have not been implemented since the project was cancelled without starting.

Additionally this technology could be also extrapolated to other irrigated agriculture areas not associated to wetlands. The technology developed in the frame of this project would be useful to more efficiently manage water in irrigation in salt-threatened lands. Many irrigated vegetables and fruit trees are sensitive to low levels of soil salinity, which threatens the sustainability of agriculture in these areas. Although there is not a general agreement about the exact extension of salt-affected areas in Europe, a good estimate was made by Szabolcs (1996), who indicated that 50 millions of hectares are more or less salt-affected. More recently, a European report elaborated by the Joint Research Centre (Jones et al., 2012), and published by the European Environmental Agency (EEA) has identified Italy, Hungary, Romania, Spain, Greece, France and Slovakia as the countries with more land affected by human-induced soil salinization. In all these countries, the system that is going to be developed in this LIFE project could be fully adaptable to improve the irrigation practices for protecting the soils against salinization.

The irrigated agricultural areas non-associated to wetlands but salt-threatened, and which could benefit from the transference of this technology are characterized by the following features:

- The use of low quality irrigation water coming from non-conventional supplies such as reclaimed wastewaters, saline groundwater, etc.

- The growth of salt-sensitive crops, i.e. many vegetables and/or fruit trees, in which a precise control of soil salinity is advisable.

- The existence of restrictions to drainage.

Green public Procurement

In light of the Green Public Procurement recommendations and to the extent that an appropriate market survey and circumstances will allow to do that, all the partners involved in the SMARTAGROWETLANDS II project will include environmental considerations in the process of purchasing and procurement of works, goods and services, moreover promoting sustainable behaviours among beneficiaries, participants and stakeholders. See a detailed work-plan under the Efforts for reducing the project's Carbon Footprint section.

SOCIO-ECONOMIC EFFECTS OF THE PROJECT

The socio-economic effects of the project implementation will be diverse and multiple. The main socioeconomic benefits are related to the conservation and improvement of the wetland habitats for birds, fish, amphibians, and the promotion of agricultural products differentiated to other regions.

In these Mediterranean regions, where there is scarcity of water for irrigation, to provide a tool for the integrated control of water amount and quality will be useful for the optimization of the irrigation rates, while saving water, adequately fulfilling the crop water requirements and keeping the soil salinity below harmful levels. This will reduce the abandon of agriculture fields, fixing population to these areas and avoiding the desertification. At the same time, it will add even more value for the trade of the local crops, which already present distinct characteristics because of the moderate soil salinity. Specifically the contribution of the moderate soil salinity to the improvement of fruit quality by reducing fruit size, and increasing total soluble solids and sugar contents, has been recognized for years. The Ravenna province and the farms of the Agrisfera co-operative do not have typical

agricultural products coming from the wetland areas, but a little more North, in the agro-wetlands of Ferrara province, the navigable channel Ferrara-Porto Garibaldi marks the Southern limit of the typical production area of the “Rice of the Po Delta”, cultivated in the wetlands of the provinces of Ferrara (Emilia-Romagna region) and Rovigo (Veneto region). The “Rice of the Po Delta” is an I.G.P. (Indicazione Geografica Protetta –Protected Geographic Indication). Its peculiarities are the high protein content, the large size of the grain, the resistance to cooking and the particular aroma due to the cultivation on brackish soils.

The cultural association “Erbe Palustri” instead, of the Bagnacavallo town (Ravenna), enhances traditional craft products obtained using plant material of wetland environments.

Additionally, this project will help to promote and develop craft industries of food processing, because the specific products yielded in these agro-environmental systems will carry an extra environmental value. A quality mark identifying the products and services of these areas will imply an added value to them.

Besides, the wetland habitat conservation associations stimulate exploration of innovative approaches to identify which services, beyond food ones, can develop markets and provide such concrete benefit. Particularly, the wetland conservation local associations, the Ravenna province and the Park of the Po Delta promote and disseminate rural tourism in all its aspects: agro-tourism, tourism bird watching, hiking through routes, etc. One of the farms of the co-operative Agrisfera (the Augusta farm), where the field experimentation of this LIFE project will be performed, is an agro-touristic farm which organizes naturalistic excursions in the bordering SCI/SPA Valle Mandriole, Punte Alberete.[AL1]

In this sense, this LIFE project does support objectives set out in the Commission Communication COM(2010) 352 of 30/6/2010, "Europe, the world's No 1 tourist destination-a new political framework for tourism in Europe", which promotes the diversification of tourism services and more in concrete agro-tourism in protected natural sites, especially of the “Natura 2000” network, as a part of the study area of this project was declared.

EFFORTS FOR REDUCING THE PROJECT'S "CARBON FOOTPRINT"

In this project we will make efforts for reducing the carbon footprint considering the following measurements: **1 In the office**

- Use of less paper as possible. It will be replaced by the email for internal and external communications.

- In case of needed use of paper we will use both paper sides or reuse already printed documents to take notes, drafts print, etc…

- We will set ‘duplex’ as the default preference for all the office printers and copiers. By printing on both sides of a sheet paper usage will be reduced by as much as 50%.

- Use printing equipment, faxes and scanners employing regular paper and no thermal paper that is not recyclable.

- Print whenever possible in black and white, avoiding inks colour consumption.

- Application of the three “R” to the residues produced in the office. Reduce, reuse and recycle all the office products (printer toners, paper, plastics, etc.).

- Regulate the cooling and heater system to comfort temperatures of the offices and labs. The office thermostat will be adjusted to 21 ºC in winter and 26 ºC in summer.

- Use of computers and screen displays employing energy saving systems, and turn off computers when they are not used.

2 In the laboratory

- Ensure that all the residues produced in the lab are disposed in the adequate container to be externally processed.

- Use of sustainable chemistry principles whenever possible.

3 For transportation

- Group the staff shift in one vehicle, whenever possible.

- Concentrate field routes for several days, staying in hotels, avoiding car use.

- Plan the field surveys and networks installing optimizing the routes to reduce the travelled distance.

- Group several activities (surveys, meetings, data requests, field routes) in the same day.

4 In meeting events and workshops

- Use telematic communication between partners or stakeholders, wherever possible.

- Considering all the measurements to reduce the carbon footprint detailed in the document published in 2008 by The Environmental Council: "The Environment Council Briefing Paper on Sustainable Events".

- For meeting participants will be recommended hotels based on their sustainability performance, and in your logistics notes.

Green Procurement

In light of the Green Public Procurement recommendations and to the extent that an appropriate market survey and circumstances will allow to do that, all the partners involved in the Life SMARTAGROWETLANDS II project will include environmental considerations in the process of purchasing and procurement of works, goods and services, moreover promoting sustainable behaviours among beneficiaries, participants and stakeholders.

In the framework of project administrative and management activities particular attention will be focused on procurement planning to assess needs, define an appropriate budget and analyze the market to have a better idea of available products/services meeting sustainability criteria.

Particular emphasis will be placed upon these aspects:

- the identification of suppliers (i.e for paper, stationery, catering services) having an environmental management system such as EMAS, ISO 14001 or equivalent

- phrasing tender specifications (or technical specifications) providing both detailed information on the functionality, quality and other characteristics (e.g. packaging, disposal, etc.) of the product/service to be purchased and a set of minimum environmental requirements which all bidders must meet;

- identification of criteria for sourcing (or pre-selecting) suppliers, vendors and manufacturers in order to assess the technical and professional qualifications of vendors to produce and/or supply the requested products. As sustainability criteria will be part of the subject matter and/or the technical specifications, sourcing criteria will be included that assess the sustainability performance of bidders to ensure that only bids from 'eligible' companies are considered in the evaluation stage; - identification of evaluation criteria to evaluate and compare the bids received which meet the minimum specifications. In sustainable procurement, it is essential to indicate that the contract will be awarded to the offer that provides “best value for money”. Evaluation criteria will evaluate the performance of a bid both in terms of price and other criteria, such as environmental performance. The tender documents published will clearly set out the various evaluation criteria that will be used to evaluate bids (such as price, technical quality, environmental quality, etc.) as well as the weight in percentage terms allocated to each aspect. Doing so, evaluation criteria can be used to encourage higher levels of sustainability performance than those demanded in the specifications, without risking significant increases in cost

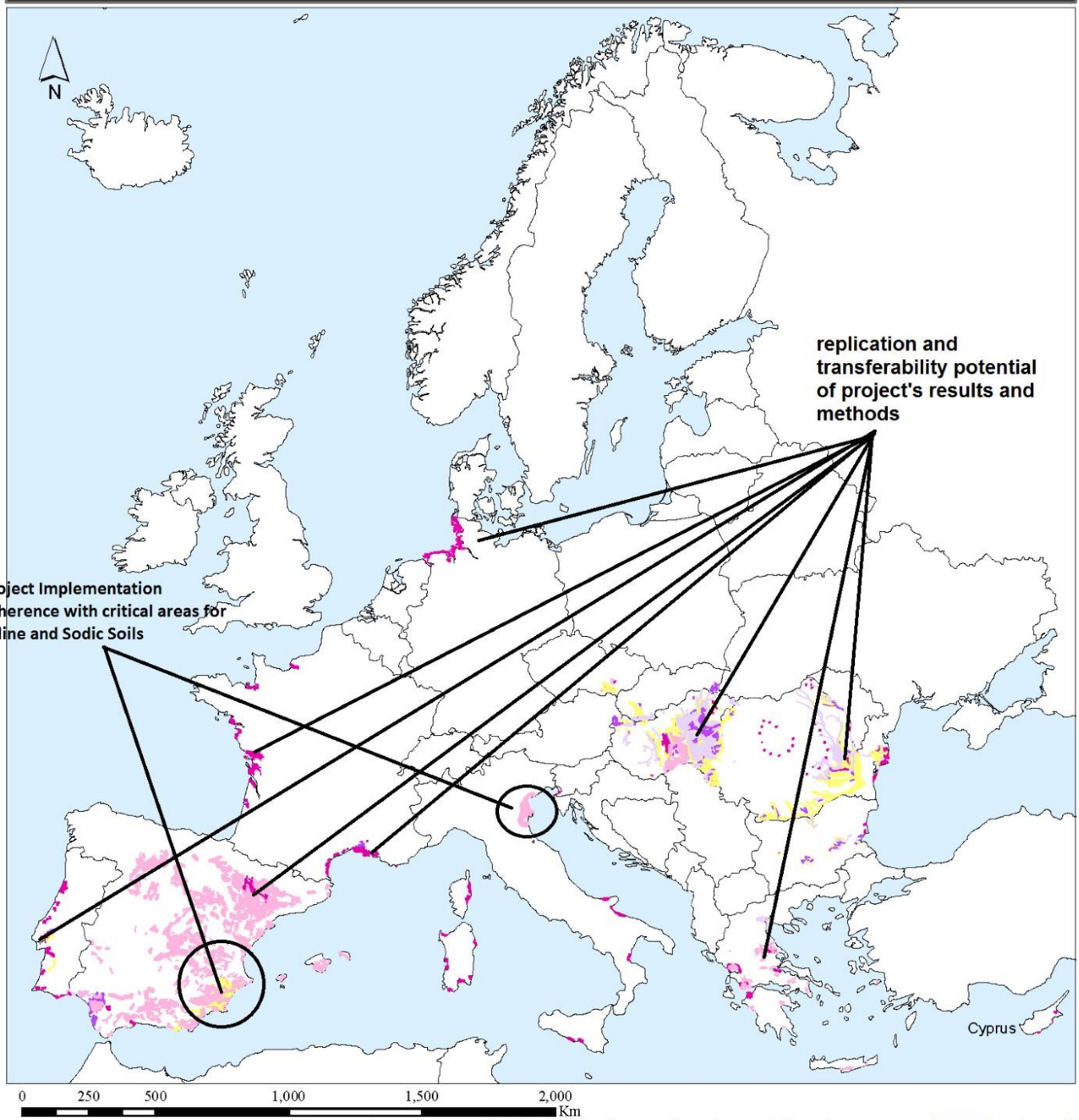
- inclusion of environmental considerations in the contract award process also including sustainability criteria in the contract clauses, if they are not included in other sections of the

tender or technical specification, with reference to penalties for non-compliance with the specifications or for cases where a supplier has provided a false written guarantee;

- purchasing of at least 65% of recycled paper and paper products;
- purchasing of at least 65% of eco-labelled stationery, consumables and computer equipment/system

Name of the picture: Figure 7 - Similar wetlands affected by agricultural activities across European

Saline and Sodic Soils in European Union



Saline and Sodic Soils

Legend

- Saline > 50% of the area
- Sodic > 50% of the area
- Saline < 50% of the area
- Sodic < 50% of the area
- Potentially salt affected area

The map shows the area distribution of saline, sodic and potentially salt affected areas within the European Union. The accuracy of input data only allows the designation of salt affected areas with a limited level of reliability (e.g. < 50 or > 50% of the area); therefore the results represented in the map should only be used for orientating purposes.

JRC EUROPEAN COMMISSION

Institute for Environment and Sustainability

MAP INFORMATION

Spatial coverage: 27 Member States of the European Union where data available.

Pixel size: 1km
Projection: ETRS89 Lambert Azimuthal Equal Area

Input data - source
Soil data - European Soil Database v2
1:1.000.000 scale Map of Salt Affected Soils in Europe (Szabolcs) 1974)

BIBLIOGRAPHIC INFORMATION

Tóth et al. (2008) Updated Map of Salt Affected Soils in the European Union
In: Tóth, G., Montanarella, L. and Rusco, E.(Eds.) Threats to Soil Quality in Europe
EUR 23438 – Scientific and Technical Research series
Luxembourg: Office for Official Publications of the European Communities p.61-74

Digital datasets can be downloaded from http://eusoils.jrc.ec.europa.eu/

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STAKEHOLDERS INVOLVED AND MAIN TARGET AUDIENCE OF THE PROJECT (OTHER THAN PROJECT PARTICIPANTS)

Stakeholders:

The main stakeholders of the project are:

- the Regional Authority (Regione Emilia Romagna, with the relevant authorities on water, regional basins and canals, environment, agriculture) (see letter of support)
- the Associazione Nazionale Consorzi di Bonifica (ANBI) –Regione Emilia-Romagna (URBER)
- the Authority of the Comunidad Valenciana
- the Municipalities of Ravenna, Ferrara and the other towns in the area (letters of support)
- the Municipalities of Valencia, Alicante, Elche in Spain and the other towns in the area
- The community of farmers (and their associations) of the drainage and irrigation area around Comacchio (Ferrara, Italy) (support: Promosagri)
- WWF Italy and other ONGs linked to EU environmental themes
- the Consorzio di Bonifica di II grado per il Canale Emiliano Romagnolo (see letter of support)
- farmers (and their associations) along the Adriatic coast and around salty wetlands (Ravenna “Piallasse” and Cervia salt-works)
- farmers (and their associations) of the Comunidad de Regantes de Carrizales
- ***La Comunidad de Regantes de Carrizales (where the original project LIFE13 ENV/ES/001125 should have taken place if it was not cancelled) see letter of Support***

The community of farmers will also be an important target for the project’s audience. The realization of the demo plant will allow to involve Basin Authority (directly involved as a Partner) and Regional authority in general (agricultural and environment departments) both in the design and implementation phase. The decision about where to realize the demo plant has already been anticipated, involving the farm owning the fields (partner), thanks to a continuous work during 20132014 , in the proposal's drafting phase, with the selection of the area of Agrisfera.

The results of the project, thanks to the dissemination activities, will be shared among the regional stakeholders and beyond in both Countries (Italy and Spain).

The replicability of the action, in both Italy and Spain, will start from this stakeholders’ involvement (especially with Regional authorities) and dissemination tools (other regions, communities with similar environmental problems).

Target groups:

Target groups of the projects, especially through dissemination activities (D1 and D2 actions), will be mostly farmers and their associations, and, more comprehensively, the farmers, conservationists and irrigation managers of Mediterranean areas in which the agriculture is developed in the surroundings of wetland ecosystems.

Although the project is specifically designed for agro-wetland areas, since the soil salinization is a process tightly associated to irrigation at large, this technology may be fully extrapolated to whatever irrigation area in Mediterranean Europe. Therefore, the target group is extended to whatever irrigation association, farmer association, and large agricultural exploitation located in the arid to semiarid Mediterranean Europe.

Together with the dissemination, the training activities represent the essential tool to transfer the knowledge on the DSS and WSN systems and its benefits to the end-users. The farmers involved will learn about the functioning and advantages of using such a system allowing them to adopt the new tool for irrigation management. More important from a dissemination point of view, additional farmers outside the Agrisfera community will be invited to learn about the benefits of the AGROWETLAND system in order to widen the impact of the project. Networking with other Life projects will also work in this direction, with their possible involvement in the training courses and other dissemination activities.

The project activities will actively be disseminated to target groups at national and European level as part of Action D.1, and they will also be included in the training activities of D.2 (both classroom and on-line training, and the filmed training session that will be made available on-line). Moreover, target groups beyond those at regional level will be approached to trial the outputs of B3 action, also thanks to the relevant stakeholders involved as potential multiplier.

Target groups from national and European level, in fact, are considered to be growers from other wetland regions across Europe, their associations and groupings, institutes and universities from Spain and Europe involved in crop science, water management, environmental protection, policy makers and agricultural ministries, standardisation and environmental bodies and other industry stakeholders from the food supply chain and water management and environmental sectors, among others.

The following national and European bodies will be targeted (among others- this is not an exhaustive list) as potential multipliers , widening the project’s target group:

- Wetlands International
- European Vegetation Survey - a working group of the International Association of Vegetation Science (IAVS)
- WetVegEurope
- European Environmental Agency (EEA)
- Vegetation Science Group of Masaryk University, Brno
- European Crop Protection Association
- Association of Friends of the Southern Alicante Wetlands
- Constructed Wetland Association
- Centro Etnografico della Civiltà Palustre (Villanova, Ravenna, Italy)
- Global Wetland Technology
- European Society for Agronomy
- EUROPEAN SOCIETY FOR SOIL CONSERVATION
- European Rice Federation
- European Agriculture Association
- IRIDRA (Italy)
- JOINT RESEARCH CENTRE – Institute for Environment and Sustainability
- SOCIETAD ESPANOLA de la CIENCIA del SUELO
- SOCIETA’ ITALIANA della SCIENZA del SUOLO (SISS)
- SOCIETA’ ITALIANA di AGRONOMIA

- WATER MANAGEMENT INSTITUTE of Lithuanian University of Agriculture (Kedainiai, Lithuania).
The main diffusion activities planned during and after the project development will be: • Design of a website to show the main results and activities developed in the framework of the project. • Organization of online and classroom workshops aimed at training irrigation advisors on the handling of the system. • Publication of notes through the project website, and also brochures, and informative articles in agricultural and irrigation professional newsletters and journals. • Communication to national and international congresses of professionals about the results obtained during the development of the project. • Organization of meetings in local forums to inform farmers, irrigation managers and agricultural engineers, and other extension staff of the benefits of using this technology. • Organization of guided field trips in the pilot areas under an expert guidance. • Promotion of the web-system in technological and agricultural fairs.

EXPECTED CONSTRAINTS AND RISKS RELATED TO THE PROJECT IMPLEMENTATION AND MITIGATION STRATEGY

The initial, technical duration of the project was considered to be of 37 months.

6 months have been added for administrative and coordination risks extending the whole duration to 43 months (2 months to include the full involvement of specific partner with low experience in European projects – see here below risk n° 3 - and 4 months for administrative issues, see here below risk n° 7)

Another extra buffer of 3 months has been added at the end of the project (total 46 months) to allow a contingency plan for actions B6 and D2 that will end , in this way, 3 months before the end of the project (see here below risk n°8)

The main constraints of the project implementation are:

1.- The low income of local farmers usually hampers the implementation of similar sensor-based irrigation technologies. This is because these technologies are perceived as expensive, and only affordable to the irrigation management of large export agricultural exploitations. So we consider the implementation of this system in this context will be highly appreciated. Besides, in order to reduce the implementation costs, an evaluation of the area aimed at minimizing the number of sensors and nodes without losing information quality has been designed in the framework of the first action. Additionally, based on the Consortium's previous experience, the most cost-effective sensors will be purchased.

Constraints, risks – contingency planning and time path: the constraint is related to the budget and technical solution, no influence on time planning.

2.- By ignorance some stakeholders could have no interest on the system. We will get some of the stakeholders involved in the project as associated beneficiaries in order to sort out this problem. This is the case of the Agrisfera co-operative farm. They in turn, could act by themselves as promoters to other stakeholders.

Constraints, risks – contingency planning and time path: the constraint is related to the stakeholder involvement and technical solution, no influence on time planning.

3.- Most of the stakeholders involved in the proposal are not used to European bureaucracy. The coordinator will give support to those unexperienced partners in fulfilling all the EC requirements.

Constraints, risks – contingency planning and time path: AGRISFERA will be supported by UNIBO. A contingency buffer, adding 2 months to the original 37 months has been included.

4.- The outdoors installation of electronic equipment always implies the risk of steal. Based on our own experience and that of the farmers, this risk will be minimized by selection of the most appropriate sites in terms of security for installation of sensors, nodes and gateways and dataloggers. Other measures may involve small solar cells and reduced visibility sensors and devices, secure moorings to civil building structures and proper signaling to avoid human contact such as high voltage, chemical or biological hazard.

Constraints, risks – contingency planning and time path: the constraint is related to the technical solution, no influence on time planning.

5.- Some local farmers, and also irrigation managers, have very little background on the handling of this kind of technology. A user-friendly and intuitive graphical interface will be designed for the decision support system in order to ease its handling. Additionally a user's manual, and an online help will be incorporated into the web-system to assist the potential users.

LIFE15 ENV/IT/000423 - B5

Constraints, risks – contingency planning and time path: the constraint is related to the technical solution, no influence on time planning.

6.- Electronic equipment in wet environments suffer the risk of being damaged. To this end, electronic equipment will be encapsulated within water-proof cases and joints such as three piece systems, and it will fulfilled the IP67 and 68 international protection ratings for mechanical casings and electrical enclosures.

Constraints, risks – contingency planning and time path: the constraint is related to the technical solution, no influence on time planning.

7 – Administrative issues - Risk of slow accomplishment of the assigned activities by project teams and partners, including delay in achieving milestones, delaying project deliverables and objectives.

Constraints, risks – contingency planning and time path: the risk is low but can not be underestimated. Effective management procedures will be applied to timely intercept problems. Partners progress will be monitored to detect any delay at early stages. A buffer of 4 months has been included (together with the buffer of other 2 months mentioned at point 3 concerning the involvement of the private partner Agrisfera), leading the project at a duration of 43 months.

8 - Project's end, final replicability actions and monitoring. Actions B6 and D2 will have their results and deliverables achieved near the end of the project. Delays in achieving the results could be dangerous because of the project's end.

Constraints, risks – contingency planning and time path: a 3 months buffer, including all the Spring Season is considered (together with the previous buffers of 2+4 months during the whole project) sufficiently reasonable to allow a smooth end of the project. AGROWETLANDS II duration has been extended to a final duration of 46 months. Budget has been increased accordingly.

9 - Possible delay of about 2-3 months in the authorization process for access to the protected areas and sampling (Sub-action C.2.1).

Constraints, risks – contingency planning and time path: to avoid the risk, UNIBO will request immediately the authorizations, during Summer 2016, before the project beginning. In case of delay, the 6+3 month buffer added to the whole project for administrative problems and project's closure should cover the risk. No budget increase should be necessary for this particular risk.

LIFE15 ENV/IT/000423 - B6

CONTINUATION / VALORISATION AND LONG TERM SUSTAINABILITY OF THE PROJECT'S RESULTS AFTER THE END OF THE PROJECT

Which actions will have to be carried out or continued after the end of the project?

The system implemented during the project will be still operative after the project's end. In particular the Agrisfera farm will continuously monitor and evaluate the salinity status in the pilot areas thanks to the features of the system itself.

UNIBO will update the software managing the system at his own costs.

Periodical ordinary maintenance of the field installation of sensors, nodes and dataloggers will be performed by Agrisfera at his own costs. Hardware repairs/substitution will be performed by OSV at cost price.

Finally, the project's Website will run for at least 5 years after the project's end. Throughout the project, the website will be organized, designed and regularly updated. Upon completion of the project, the website will be updated introducing new information in the sections of the website such as events and news of interest, recent information, guided field trips, environmental activities as birds banding and watching, etc.

More important, the actions undertaken in Spain during action B6 will be the starting point for a first replication phase that will be fully developed after the project's end. At the same time, other farm, through the displayed training actions should be able to start, replicating and applying the demonstrated features and techniques for DSS and WSN systems.

At the end of the project, the server to implement the SMAR AGROWETLANDS II system (acquired by IRIS) will be left at Agrisfera disposal (being not recoverable) for the future use in field.

How will this be achieved? What resources will be necessary to carry out these actions?

The main action for monitoring and evaluation of the salinity status after the end of the project can be automatically performed by the own implemented system. This system will be designed to store information of the main variables that represent the salinity status of the agro-wetland ecosystem such as the irrigation water volume, and its electrical conductivity, soil salinity at several soil depths, piezometric level, and volume and electrical conductivity of the drainage water. Periodically the irrigation managers could remotely access the web-system to know the levels of water flow, salinity and soil salinity. With this information we can monitor the evolution of salinity in soil and in drainage water and evaluate the level of salinization of the ecosystem. All this information can be input in the project website with elaborated statistics of the main variable measurement with the system. Since this action does not generate any cost can be done without any funding.

More recent web system versions can be installed remotely to update the software that manages the system. This can be done by Unibo and no extra funding is necessary for this action.

Using the system, the irrigation managers will be able to check the functionality of the sensors. In case of malfunction of the sensors, the irrigation association personnel can reinstall the sensors. The most robust sensors will be installed to minimize this problem.

The replicated activities, once the original action has been fully demonstrated, will be able to find their way to the realization thanks to the private funds of the farmers, but especially with the regional European Agricultural Fund for Rural Development (EAFRD), thanks to the potential of the demonstrated instruments, able to allow a better production and soil and water management, meeting several priorities of the regional programmes on rural development.

In particular, Emilia-Romagna Region PSR , in the priority 5, includes the support to already demonstrated solutions for carbon/soil/water related intervention in agricolture and for this reason will be used both to carry the future actions and for replication activites.

To what extent will the results and lessons of the project be actively disseminated after the end of the project to those persons and/or organisations that could best make use of them (please identify these persons/organisations)?

LIFE15 ENV/IT/000423 - B6

Before the project's end, the project board will establish the AFTER LIFE communication plan for the sustainability of the project's results. This plan (being an important section of the AFTER LIFE PLAN , see sub-action E.1.3) will include a summary evaluation of the project's activities and a specific section for the future dissemination activities, aiming at replicability, of Agrowetlands II.

The people or organizations that have been so far identified as targets for dissemination activities are:

- Local population: NGOs, individual citizens, farmers, conservationists, in Italy (Emilia Romagna Region), and Spain (Valencia, Alicante, Elche)
- Agricultural sector: irrigation associations, agricultural co-operatives, extension personnel.
- Local and/or regional governments in Italy and in Spain (Emilia Romagna Region, and Comunidad Valenciana)
- Managers of similar Natural Parks to the pilot areas in both Italy and Spain /Regional Park of the Po Delta, Parc Natural de l'Albufera, Southern Alicante wetlands, wetlands of Elche, Italian and Spanish sites of the Ecological Network Natura 2000.
- Scientific and academic community.
- Technology companies focused on agricultural activities.

While the general public, especially the regional and local one, will be addressed through the website and dissemination materials, an important part in the dissemination strategy will be played by: - Networking

- Training actions
- Participation to multipliers events

Networking with other LIFE projects will expand the borders of the projects outside the area of the action, sharing best practices and making use of them for future implementation.

The training action will be the basis for both a concrete maintenance of the results on site and an implementation beyond the limits of the demonstration plant at local and regional level.

The participation to multipliers events like fairs on agriculture technology, water and soil management, together with the support of dissemination material and website, will allow a better sharing of the results of the projects, showing the possibilities to implement the WSN-Web system solutions in other regions and with which potential results and costs.

LIFE15 ENV/IT/000423

TECHNICAL APPLICATION FORMS

Part C – detailed technical description of the proposed actions

LIST OF ALL PROPOSED ACTIONS

A. Preparatory actions (if needed)

B. Implementation actions (obligatory)

B1 Environmental database construction, modelling of water and salt flow in the pilot area

B2 Development of a new model of soil and water salinity for irrigation recommendation

B3 Development and implementation of web-control system and user interface for irrigation management

B4 Development of an integrated WSN modules to monitor soil and water salinity, and water-table level for the sustainable irrigation

B5 Field installation of wireless sensor network in pilot area

B6 Field application of SMART-AGROWETLANDS II and evaluation of its effects on crops

C. Monitoring of the impact of the project actions (obligatory)

C1 Monitoring the Project’s impact on salinity status in agro-wetland pilot area

C2 Monitoring the Project’s impact on the natural and semi-natural habitats, plant communities and biological quality of water and assessment on socio-economic impact

D. Public awareness and dissemination of results (obligatory)

D1 Communication Pack and Dissemination Activities

D2 Networking, Training activities and Dissemination Events - Replicability

E. Project management (obligatory)

E1 Project management, monitoring and After Life Plan

DETAILS OF PROPOSED ACTIONS

B. Implementation actions (obligatory)

ACTION B.1: Environmental database construction, modelling of water and salt flow in the pilot area

Description and methods employed (what, how, where, when and why):

The first project action, subdivided in the sub-actions B1.1 and B1.2, aims to create the framework for the description of the pilot area as well as of the water and salt flow in it.

Sub-action B1.1 will start at the beginning of the project, whereas sub-action B1.2 will start at the conclusion of the first field surveys, in the second trimester of the 2017. After this date, the two sub-action will continue together, till the end of September 2017.

The description of the area will be reached through:

- the environmental characterization of the pilot area, by field surveys in a representative number of plots, for several static and dynamic parameters of the soil, channel water quality, water-table depth, crops and irrigation management (sub-action B1.1);

- the construction of an environmental database, supported by a Geographic Information System, containing:

o all above data gathered in the survey plots and the extension by a geostatistical approach to the whole pilot area o flow monitoring at selected sections and modeling results.

It is felt useful to decide reasonably a first set of location where to perform sampling and to decide locations a second set later in the project. Numbers are respectively 8-12 for level-temperature-salinity monitoring stations(See Figure B.1.1), and 30-20 for plots data.

B1.1 Environmental characterization of the pilot area

Soil sampling and analysis will be performed 50 geo-referenced plots, covering approximately 600 ha of the Agrisfera area. Measurements in each plot include the estimation of the soil saturated hydraulic conductivity and the soil bulk density in the depth intervals 0-10, 10-30, 30-60, 60-90 cm. Portable sensors will be used to take measurements of apparent electrical conductivity, and soil water content at the selected depths. Samples of soil will be collected at the same depths of the field measurements, for chemical (pH, organic matter content, calcium carbonate equivalent) and physical laboratory analyses (texture, volumetric water content at field capacity and at permanent wilting point, bulk density, electrical conductivity at 25 °C of the saturation extract repeated on samples collected in winter and during the irrigation season).

Water flow measurements and sampling will be performed on the major irrigation and drainage channels of the pilot area. 50 flow measurements and 300 water samples, water elevation measurements and flow direction identification are sufficient to have a detailed enough information of the water quality and of salt flows inside the areas and at their border. Sampling points will be located by GPS and visited in two distinct periods of the annual hydrological cycle to outline the time evolution of the water quality and flow rate: in winter when channels are used to drain the area and during the irrigation season, from May to September. The water samples will be analysed in laboratory for electrical conductivity at 25 °C, pH, main ion and nutrient contents, chemical and biochemical oxygen demand, oxygen saturation, turbidity and alkalinity.

Monitoring of water elevation in channels and its seasonal fluctuation will be performed through 8 piezometers at channel banks(See Figure B.1.2),with 2 m windows to detect stratification if present. Piezometers will be properly distributed over the pilot area in geo-referenced sites.

Information on crops and irrigation management will complete the environmental characterization of the pilot area. The farmers of Agrisfera will be interviewed on the current cultivation practices (type of crops, irrigation rates and scheduling, water supply, fertilization, planting and harvesting dates, and other agricultural practices). All the agricultural information will be spatially geo-referenced.

B1.2 Database creation and modeling of salt and water flow

Just after the first sampling and analyses, a database will be created to store information gathered by subaction 1.1 and associated to each sector (soil, water, water-table depth, crop and agricultural practices), fulfilling the ***INSPIRE directive***. Two main types of geo-referenced information will be managed:

- Time dependent variables: soil and water conductivity, water-table depth, crop types, irrigation management, meteorological characteristics (rainfall, evapotranspiration, etc.);

- Quasi-Stationary variables: soil chemical (organic matter content, calcium carbonate equivalent) and physical parameters (volumetric water contents at field capacity and at wilting point, texture, bulk density).

In order to characterize salt intrusion dynamics in the study area and to assess its evolution in space and time, a numerical model of channel and groundwater flow as well as of solute transport will be implemented. Open source 3D finite-difference models will be used that are extensively used for simulating and predicting ground- and surface water interaction:

MODFLOW (http://water.usgs.gov/ogw/modflow/), SEAWAT (http://water.usgs.gov/ogw/seawat/).

A geostatistic methodology will be used to evaluate the spatial distribution of soil properties. The analysis will be performed following the steps: a) exploratory data analysis evaluating variable frequency distribution, b) spatial continuity assessment and modeling (i.e. fitting experimental and theoretical semivariograms accounting for flow information); c) spatial interpolation using several kriging methods; d) uncertainty analysis.

Soil chemical and physical parameters and water-table level will be analysed. The resulting maps will be integrated in a GIS framework, implementing the following data layers: crop distribution, agricultural plots delimitation, irrigation channel network, drainage channel network, irrigation system and rates, groundwater level and quality. GIS tools will be used to combine the information from these layers in order to obtain the spatial distribution of the parameters involved in the water management of the study area.

Based on the spatial distribution of the parameters listed above, on flow direction and site security, during the last trimester the more adequate sites where to place 12 further piezometer at channel banks and 20 piezometers in field will be selected, so that representative information is provided to the DSS.

Expected results:

- Environmental characterization of the pilot area through a database fulfilling INSPIRE Directive

- Numerical models describing surface water flow and quality, groundwater system and saltwater intrusion in winter and in the irrigation season

- First evaluation of soil and water salinity to be used as baseline in comparison with the status after the implementation of the SMART-AGROWETLAND system

- Determination of the optimum number of salinity sensors and their location in the farms

Indicators of progress:

- Percent of the pilot area surveyed for soil, water and crop spatial distribution

- Number of farmers interviewed

- Percent of channels modeled

- Number of selected representative sites to install sensor

Beneficiary responsible for implementation:

UNIBO

UNIBO will be assisted by the Agrisfera co-operative farm

Assumptions related to major costs of the action:

The staff will be from UNIBO and Agrisfera, composed by expert professors and technician in the following sectors: agronomic, engineering, hydraulic engineering, pedology, technicians

Compliance with the eligibility conditions in the grant agreement (Art. II.19.1) will be respected indicating real costs based on real salaries and social charges, as indicated in the previsional table in Annex RP12 attached to the proposal.

All travels are concerning UNIBO staff (being Agrisfera already *in situ*) Travels have been reduced as much as possible, but are necessary for the high number of surveys foreseen in this action.

Equipment (*4.200 €)*

4.200 € for a server for hydraulic modelling, e5 xeon. This server will be also used in action B2 and B6. Only 50 % of the cost has been considered as eligible. Total eligible cost is 2.100 €

Consumables (9.250€)

1.800 € : Portable sensor for on-site soil salinity measurements

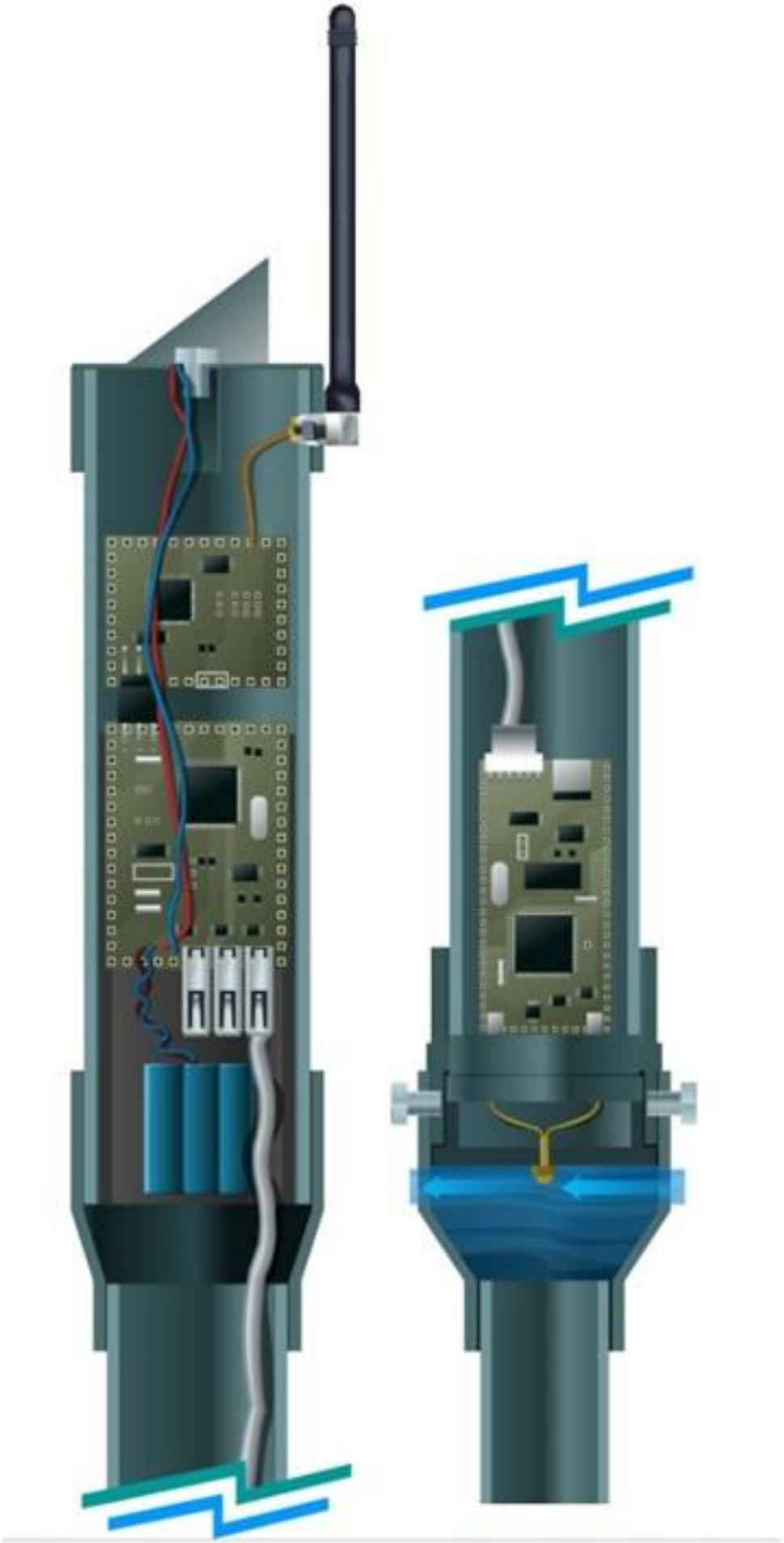
3.600 € : 8 piezometers with 2 m windows

1.800 € : Chemical reagents for water laboratory analysis of 300 samples at 6 €/samples

1.600 € : Chemical reagents for soil laboratory analysis to determine of soil organic matter, carbonates, ph and electrical conductivity, and water salinity for 200 soil sampling at 8 €/ sample

450 € : Plastic bags, plastic pots and other field material for taking samples.

Name of the picture: Figure B1.1 - Salinity Sensor: top part (on the left), bottom part (on the right)



Salinity Sensor: top part (on the left), bottom part (on the right)

Name of the picture: Figure B1.2 - Piezometer and salinity sensor at a channel bank

Piezometer and salinity sensor at a channel bank

B1's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
Geo-database with the main soil, climate, watertable, water salinity, water flow parameters	06/2017

B1's PROJECT MILESTONES

Milestone name	Deadline
Initial spatial evaluation of soil salinity, water-table level, crops, watr flow and salinity in irrigation and drainage channels within each sector of the pilot area	07/2017

Spatial delimitation of sectors for water management	09/2017
Definition of spatial localization of sensors within the sectors	10/2017

B. Implementation actions (obligatory)

ACTION B.2: Development of a new model of soil and water salinity for irrigation recommendation

Description and methods employed (what, how, where, when and why):

B2 Action represents the core of the project, since during this action a model for irrigation scheduling will be, for the first time, developed and implemented over the pilot area, with the purpose of saving irrigation water and control salinity. The performance of several existing models estimating leaching has to be compared to define a model tailored to specific local conditions. The calibration and validation of the model will be performed off-line by using the data sets collected during B1 Action. The model will be further tested during the entire B6 Action by employing the data measured by the sensors.

Due to the novelty of the irrigation management system to be developed within this action, where for the first time water and salinity parameters are going to be integrated into the decision making system, there is an amount of risk involved. We have foreseen enough time (2 years, from October 2016 to September 2018) to be devoted to this action. There is sufficient knowledge in the university team to be sure that this can be successfully achieved.

Tasks:

A model will be created to adjust the irrigation doses and times in each field within the pilot area depending on crop water requirement, crop growth, soil moisture and salinity as well as irrigation water salinity and water-table depth. Compared to the WaterBee Soil Moisture model, the new model will take into account three additional factors:

i) the soil salinity ii) the salinity of the irrigation water iii) the

water-table depth and the drainage water salinity and flow

The model will work coupled with a database where all the information on crop and soil of each plot will be saved. These relevant data for irrigation scheduling will be assembled from available information and new data obtained in the first action (B1). The soil properties in unvisited plots will be estimated by means of geostatistical analyses as described in previous action (B1). For crops the information will be species, age (trees) or planting date (vegetables), basal evapotranspiration coefficients, maximum canopy cover and maximum rooting depth. For soils the information will be hydraulic properties per soil depth such as stone percentage, bulk density, and the water content at field capacity and permanent wilting point.

The model will receive instantaneous input from the field sensors, as described in the fourth action (B4), located in representative places selected as indicated in B1 within the pilot area. The crude soil data will be apparent dielectric permittivity and apparent electrical conductivity at different soil depths. From these the model will assess the soil water content and soil pore water salinity using a database of calibration parameters.

Both apparent dielectric permittivity and electrical conductivity will be combined into a single crop hydric stress factor, which will be subsequently compared with critical values. The critical values will have been previously calculated on the basis of crop development stage and soil hydro-pedological properties. In case the crop is stressed according to the previous comparison, an irrigation dose will be calculated on basis the same crop and soil data as before and additionally data of irrigation water salinity, i.e. irrigation water electrical conductivity, irrigation canal flow and also water-table depth. The model will simulate the resulting soil salinity and water-table depth, and the irrigation dose will be reassessed in case either excessive soil

salinity or water-table rise resulted. This information will be available to users as irrigation recommendations (Figure B 2.1.)

Since long time, water and salinity stresses have been independently evaluated. However, it is widely acknowledged that both are linked through the soil water potential concept. That is, as water content drops or salinity rises, the soil water potential decreases. Accordingly, through the soil water potential a single combined hydric stress factor, which is something new, will be developed. To our knowledge there is no irrigation recommendation system that uses this unifying concept as calculated from the individual values of the soil water content and soil pore water salinity.

In this sense, for assessing soil water salinity UNIBO will compare the performances of several existing models in scientific literature accounting for the effects of some major physical and chemical processes. Some of these models are indicated in Table B2.1. The model, that will provide the more realistic leaching requirement estimate for the study area, will be implemented in a Web site to provide on-line irrigation management recommendations to farmers.

In establishing the working rule, the numerical model based on the MODFLOW code will be applied off-line evaluating the effects of the different models simulating soil salinity.

Results:

- A single parameter for irrigation recommendation that takes into account simultaneously both water and salinity stresses will be available. Presently we could not give critical values for this new parameter because it has not been developed yet.

- A new decision support system will be available for irrigators in the demonstration area. In upcoming years if needed it could also be adapted for localized irrigation systems to give rise to automatic irrigation.

Beneficiary responsible for implementation:

UNIBO

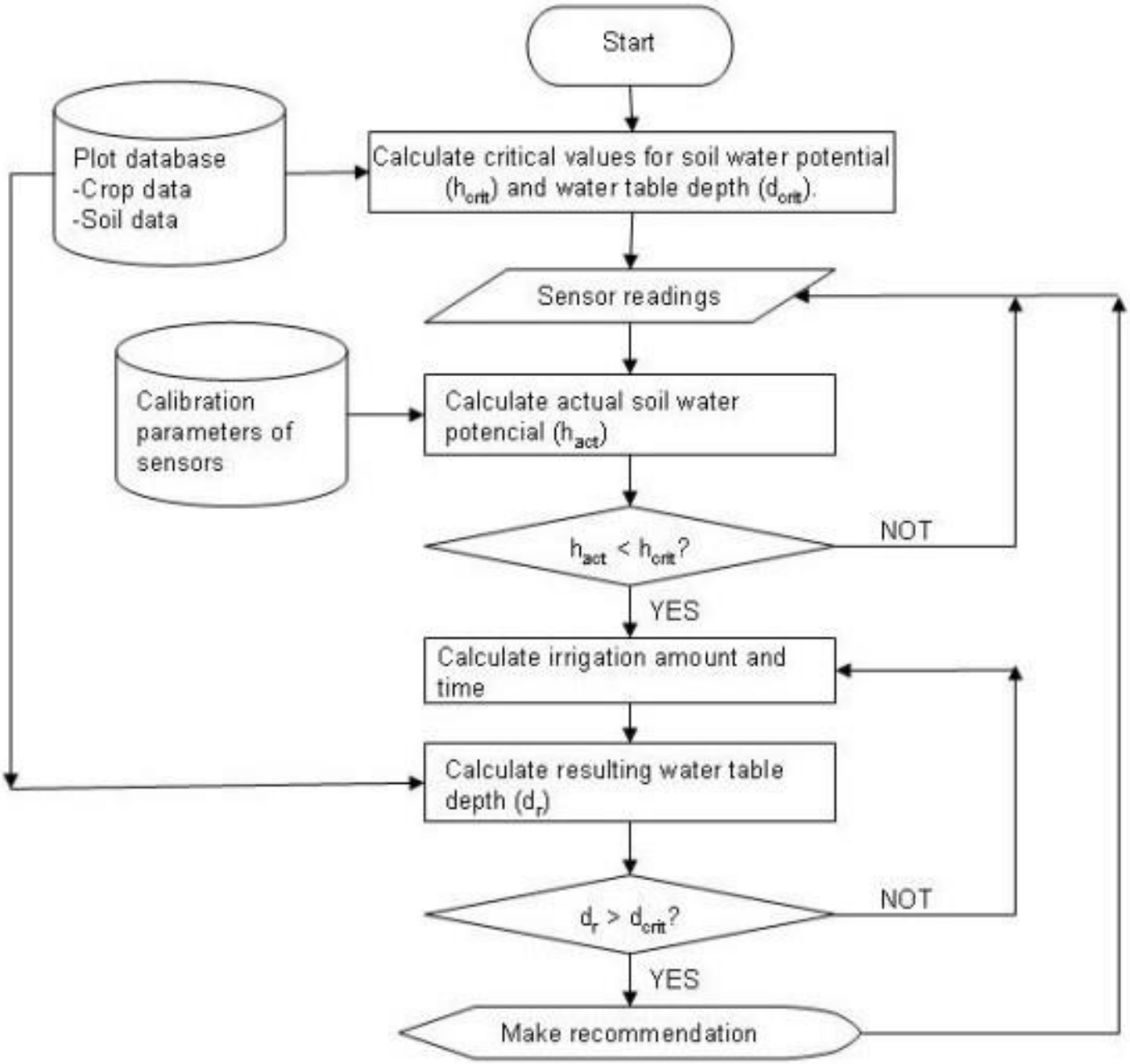
IRIS will assist UNIBO in the implementation of the SMART-AGROWETLANDS II system.

Assumptions related to major costs of the action:

Consumables (450€)

450€: 30 Pots of 50 liters each, and enought soil sustrate for experimental essay to check the new module.

Name of the picture: Figure B.2.1



Name of the picture: Table B.2.1

Table B2.1 – Models estimating leaching requirement

Leaching Requirement Model	Type of Model		Processes Included in Model		
	Steady-State Model	Transient-State Model	Salt Effects on Plant Growth and Evapotranspiration	Osmotic and Matric Effects on Plant Water Uptake	Precipitation-Dissolution Reactions
WATSUIT [1]	X				X
SALTIRSOIL [2]		X	X		X
UNSATCHEM [3]		X	X	X	X

[1] Rhoades, J.D., and Merrill, S.D. (1976). Assessing the suitability of water for irrigation: theoretical and empirical approaches. In: Prognosis of Salinity and Alkalinity. FAO Soils Bulletin, 31, FAO, Rome, Italy, 69-109.
[2] Visconti, F., de Paz, J.M., Martinez, D., Molina, M.J. (2014). Irrigation recommendation in a semi-arid drip-irrigated artichoke orchard using a one-dimensional monthly transient-state model. Agricultural Water Management, 138, 26-36.
[3] Suarez, D.L., and Šimůnek, J. (1997). UNSATCHEM: Unsaturated Water and Solute Transport Model with Equilibrium and Kinetic Chemistry. Soil Science Society of America Journal, 61, 1633-1646.

B2's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
New model code for water management linked to the geo-database	09/2017
User Manual and description of model functionality	02/2018
Report on AGROWETLANDS water management model	08/2018

B2's PROJECT MILESTONES

Milestone name	Deadline
model of soil and water salinity for irrigation recommendation (first version)	07/2017

model of soil and water salinity for irrigation recommendation (second version) checked at small scale in crops cultivated in pots	01/2018
model of soil and water salinity for irrigation recommendation (final version) checked at field scale in pilot area	07/2018

B. Implementation actions (obligatory)

ACTION B.3: Development and implementation of web-control system and user interface for irrigation management

Description and methods employed (what, how, where, when and why):

In order to successfully make data from the model developed in the previous action easily accessible to the end-users, a web control system is needed. Additionally, the direct communication interface with the farmer (or end-user) is the user interface, that is essential for the correct usage of the obtained data, therefore a thorough development of these tools is essential to ensure the usability and success of the system.

The main objective of this action is therefore to develop the web-platform software for the on-line environmental monitoring of the pilot areas managed by the Agrisfera co-operative farm. The action lasts for two years from October 2016 to September 2018; sub-action B3.2 interacts with the action B.2.

The web-platform will be based on early developments from Smart-Paddy and WATERBEE projects (Figures B3.1, B3.2, B3.3), will provide access to the geo-database developed within Action B1 and will include the irrigation management procedure developed within Action B2. Additional data acquired with the WSN developed within B4 and the local weather station will be displayed and included into the management software developed in Action B2.

The web-platform will be developed using existing open-source software solutions. The User Interface will be developed to access the entire environmental database and present the results from the data collection in an understandable and customizable form. Extensive user tests will be performed to ensure that the design meets the needs and expectations of the end-users of the system. The objective is that the final user becomes quickly familiar with the interface and can adapt the presentation of the data in the most appropriate way for his/her needs.

For implementing the software development work the SCRUM agile methodology will be applied. This methodology is suitable for developing complex applications that include hardware interfaces, SQL and NoSQL databases (MySQL, Postgres DB, Couchbase among others) and advanced and end-user friendly graphic interfaces not only for computers (using Windows Presentation Foundation) but also for smartphones (Android, iOS) or web-based access (HTML5). This methodology is developed by dedicated programming teams under a dynamic and responsive work scheme oriented to delivering client-focused tasks.

Training activities for the use of the web-based user interface will take place both for the farmers within the project consortium and for external potential users within Action D2. End-user requirements need to be channeled to the software developers since the beginning and during the whole action. This will guarantee the proper demonstration of this innovative soil and environmental monitoring tool. In order to achieve this, training and dissemination activities will be defined in conjunction with Action D1 and D2. In order to meet the needs of the end-users based on their technology habits, a smartphone version of the user interface will be developed.

Sub-action B.3.1 Web-platform design

The web-platform will be based on MVC (Model-View-Controller) concept (Figure B3.4) that isolates the database with its processing (the Model) from the data manipulation (the Controller) and the presentation of the data in web based user interface (the View).

The Model layer of the SMART-AGROWETLAND II system defines the business logic with the database and the irrigation model belonging to this layer. The main parts of this layer are developed within B1 and B2.

The Controller in this system software architecture is a piece of code that calls the irrigation model developed within B2 to get some data that it passes to the View for rendering to the client. All requests are managed by front controllers. These front controllers delegate the real work to actions that are logically grouped into modules.

The View is what the user interacts with. This layer is mainly made of PHP templates, which allow a rapid development of views, compatible with any viewing device (PC, phone or tablet) and any web browser. The view part of the SMART-AGROWETLAND II software is developed with open-source technologies.

A number of different programming languages exist for such a purpose. Three languages that are commonly used for interacting with databases on the Web are ColdFusion, Active Server Pages and PHP. There are also a
number of other languages that may work equally well for different situations including Perl, Python, JavaScript and VBScript.

Languages and structures that may be selected, following previous projects, are a MySQL database with PHP and AJAX (Asynchronous JavaScript and XML). As a data-interchange format inside the system is used JSON (JavaScript Object Notation). It is a lightweight format, is easy for humans to read and write and for machines to parse and generate.

MySQL is a relational database management system which nowadays has more than 6 million installations. It is commonly used by free software projects, is popular for web applications and acts as the database component of the LAMP (Linux operating system + Apache web server + MySQL database management system + PHP scripting language) software stack. One of the advantages of using such a development environment is that web applications can request only the content that needs to be updated, thus drastically reducing bandwidth usage and load time.

Sub-action B.3.2 Web-platform and user-interface development

In close interaction with UNIBO and the end-users of the project, the web-platform will be developed. Special attention will be taken in order to ensure compatibility with previous actions. The database from Action B1 will be enlarged and adapted to include the measurements of the WSN; the weather station and irrigation software models developed within B2 will be integrated into the web-platform system.

The design of the user interface will be based on the end-user needs with the objective of ensuring a good impact of this innovative monitoring tool in the farming communities.

The measured values with the WSN will be displayed and easily accessible in the web-site providing farmers with irrigation management recommendations that ensure a sustainable use of soil and resources in the area.

An adapted version of the SMART-AGROWETLAND II User Interface for different Smartphone platforms will be also developed.

Expected results:

A reduced preliminary version of the system will be available at month 12.
A fully functional web-platform will be ready for end-users with at least the following functionalities:

- Direct access to weather values.

- Location of all the measuring nodes in the pilot area.

- Direct access to the WSN readings (soil water content, soil salinity, soil temperature, irrigation water level and salinity, and drainage water level and salinity) and different options for data display.

- Direct access to the geo-database measured by on-site instruments.

Comparison between the different sensor readings.

- Irrigation recommendations.

- Alerts display

Beneficiary responsible for implementation:

IRIS

UNIBO and OSV will support IRIS

Assumptions related to major costs of the action:

Other costs (3.000 €)

3.000 €: Server to implement the SMART-AGROWETLAND system (not equipment, but completely not recoverable once inserted in the field) to be left at Agrisfera disposal in the filed after the end of the project

Name of the picture:

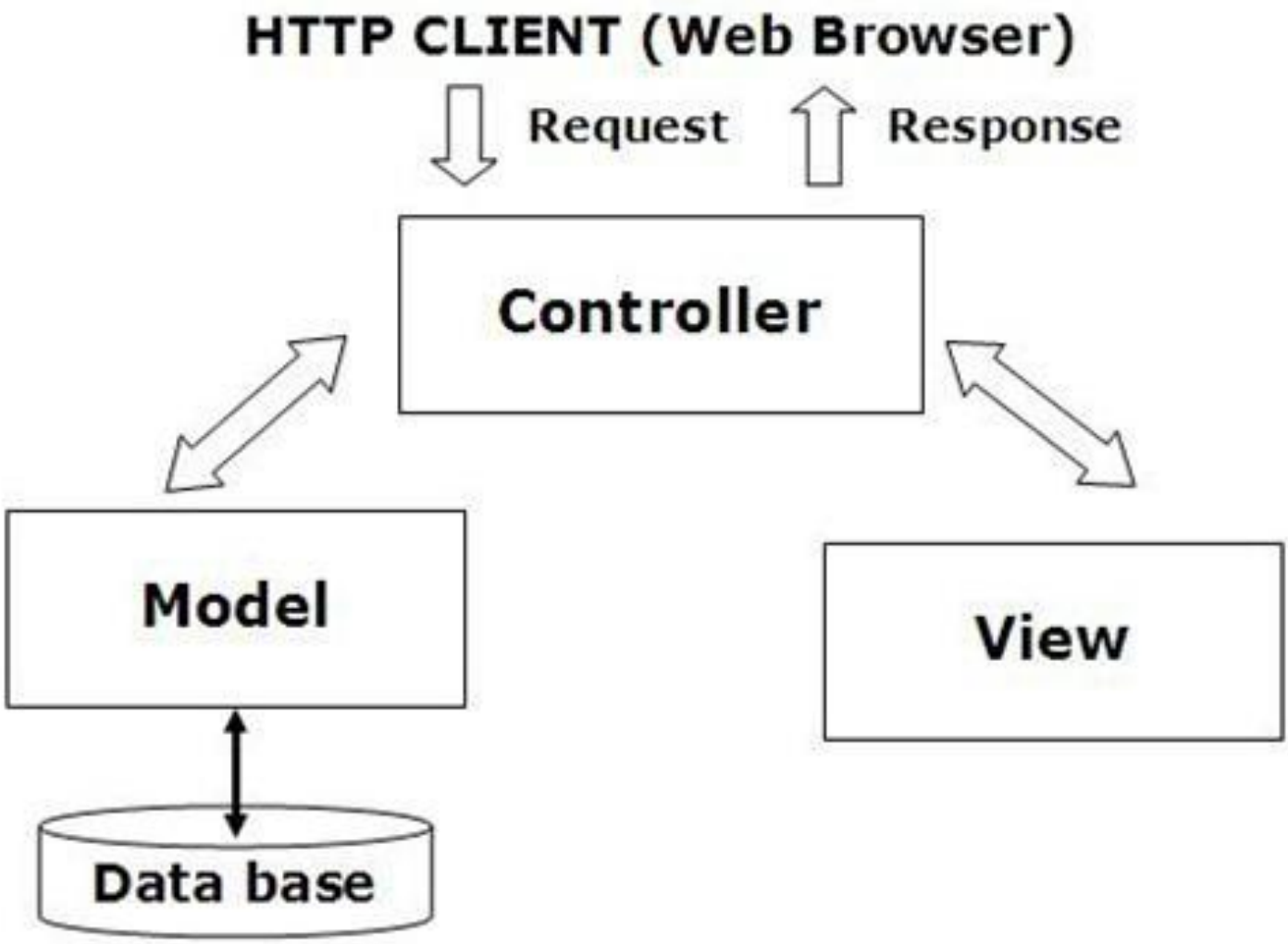


Figure B3.4 – Model-View-Controller (MVC) concept

Figure B3.3 – Screenshot of the SMART-PADDY User Interface

B3's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
Report on the Agrowetlands web-platform	08/2018
Web-based User interface user manual	08/2018

B3's PROJECT MILESTONES

Milestone name	Deadline
Agrowetland web- platform (web-control system and user interface for irrigation management) final version up and running	07/2018
Preliminary version of web control system available	09/2017

B. Implementation actions (obligatory)

ACTION B.4: Development of an integrated WSN modules to monitor soil and water salinity, and water-table level for the sustainable irrigation

Description and methods employed (what, how, where, when and why):

In order to successfully integrate in the models real data obtained from the study area it is essential to develop:

• a design of the WSN that will be collecting data from the field,
• an integrated multisensor node with all its features (power supply, communication and data storage
• a communication protocol to receive and transmit data between sensors.

The testing of the developed system will also be an essential task to ensure that the WSN:

· is performing up to the desired standards,

· is providing the necessary installation to carry out the soil and water improvement studies envisaged in later tasks.

For this reason, an innovative WSN instrument will be developed for the on-line monitoring of soil quality, water salinity and water-table level in the Agrisfera pilot area. Following the recommendations of Action B1, the required number of measuring nodes will be manufactured. Both hardware (sensors, communication and power supply units) and software will be integrated for the measurement nodes and the gateway for the transmission of data into the internet.

Within this action a heterogeneous WSN for the remote measurement of soil and water salinity and water table level will be manufactured. The WSN is comprised of four different parts, the soil measurement nodes, the water measurement nodes, piezometers for the water-table measurement and the gateway for the collection of data. A schematic picture of the monitoring system is shown in Figure B5.1.

In the WSN, autonomous sensor nodes (powered by rechargeable via solar cells battery) measure, for instance soil moisture, water table level and salinity at given time intervals and transmit the data to a central node or gateway. The gateway device is able to collect data and transmit them via internet to a remote server which hosts a web-based user interface. Farmers or any interested end-user can access the measured data from anywhere via the web-platform. Intelligent software operates with these data and provides advice to assist farmers in the decision making process (see Action B2).

The action will foresee the realisation of the following outputs:

Wireless Sensor Network (WSN) design

A heterogeneous WSN (with water quality, water-table level and soil quality sensors) will be designed for monitoring the Agrisfera area. The final number of nodes will be defined following the requirements defined within B1 in order to ensure the proper environmental monitoring of the pilot areas of the Agrisfera cooperative farm. Initial estimates are included in B4, and comprise a total of 4 gateways, 20 soil nodes and 20 water.

Based on the results of B1 and on the analysis on the dimensions and communication distance capabilities of the WSN, the final configuration will be established. Additional water sensors will be installed directly in two points of the natural and semi-natural habitats.

Water node integration

The water node for the salinity measurement will be fully integrated and manufactured based on the SMARTPADDY technology. The EC and pressure sensors, the batteries and solar cells for the power supply system, together with the wireless communication transceiver and antenna will be electronically integrated into a central printed circuit board. The mechanical encapsulation of the system will be adapted to the requirements of the precise measurement locations (drainage or irrigation channels) in the Agrisfera area and the required measurement depths.

Soil node integration

The node for the soil humidity monitoring will be based on the WATERBEE technology. Soil water content, temperature and salinity sensors (type GS3 or similar of Decagon) will be used and integrated into the proper wireless communication technologies. Some additional technical adaptations will be performed in order to integrate the system into the heterogeneous WSN.

Gateway integration

A master node or gateway device will be used as the central on-field point in the WSN network. It will act as the highest hierarchy node within the network, and will have bigger computational capabilities than the rest of the nodes.

The gateway will have different connection options to be able to transmit the information from inside the field (via RF or ZigBee) to a remote central server database via Mobile Data GPRS/CMDA or any other wireless communication channel that is available in the Agrisfera area.

The information will be sent to a central database that will serve a dual purpose: (a) to control the SMARTAGROWETLAND II system and, (b) to act as a user interface with the system.

The power supply design specifications for the gateway will be defined depending on the facilities available. A solar panel will act as the power supply and additional rechargeable batteries may be added depending on the power requirements of the final WSN design.

Communication protocol

Software that integrates the communication between the three different nodes will be developed, always keeping in mind the minimal power consumption of the autonomous sensor nodes, such that no recharging is needed within a full harvesting session.

Functional testing of the fully integrated WSN

Once all the elements have been fully integrated into the WSN, preliminary laboratory testing prior to the installation will be carried out at IRIS in order to ensure that the integrated components work at peak performance and smoothly together.

Expected results:

A reduced number of modules will be available at month 10 and the fully integrated WSN system will be ready for installation by month number 20.

The initial estimated number of nodes is 4 gateways, 20 water nodes and 20 soil nodes.

Soil quality and water level and salinity readings at fixed time intervals will be available on-line in a webbased user interface.

Solar power supply system (with maybe additional batteries) will provide autonomy to the sensor nodes during a full harvesting season.

Indicators of progress:

- Number of gateway units manufactured

- Number of sensor nodes manufactured

- Performance of the integrated WSN will be indicated via validation tests.

Beneficiary responsible for implementation:

IRIS

Both IRIS and OSV participated in the SMARTPADDY and WATERBEE projects. This will ensure that technology from both R&D projects can be fully and properly transferred into the project. IRIS will be in charge of integrating the soil and water sensor modules into a heterogeneous Wireless Sensor Network. OSV is an agricultural equipment manufacturer and will assist IRIS in the manufacturing of the system. OSV will provide the WATERBEE hardware technology.

Assumptions related to major costs of the action:

Consumables (22.950 €)

€ 3.700 PCB mounting (IRIS)

€ 2.000 salinity EC modules (IRIS)

€ 4.000 mechnical parts for encapsulation of the gateway (IRIS)

€ 9.000 mechanical parts for encapsulation of the 20 sensor nodes (IRIS)

€ 4.250 No. 1 Embedded microcomputer (ALIX) for the gateway units of the WSN in Italy (OSV)

Other Costs (5.000 €)

Altium License (2.500€)

License to IAR-Embedded Workbench (2.500€)

B4's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
Report on the integrated Agrowetland WSN, including functionality tests	04/2018
Installation and User Manual of the Agrowetland Unit	04/2018

B4's PROJECT MILESTONES

Milestone name	Deadline
Fully integrated heterogeneous WSN is ready for the installation in the Agrisfera agrowetland	04/2018
First set of sensors ready for the installation at the beginning of action B6	03/2017

B. Implementation actions (obligatory)

ACTION B.5: Field installation of wireless sensor network in pilot area ***Description and methods employed (what, how, where, when and why):***

The main aim of this action is the field installation of:

- the minimum number of sensors providing representative information for the local irrigation management;

- the weather station for monitoring of climatological conditions;

- the gateways necessary to receive all the data from the sensors and retransmit them to the central control server.

The installation of the wireless sensor network in the pilot area will permit the real-time monitoring of all parameters of interest for the salinity control.

For this reason, once the spatial delimitation of homogeneous sectors of water management has been carried out in the framework of the B1 Action, the installation and instrumentation of each sector with the minimum number of sensors providing representative information to advise the water management will be performed. In Figure B5.1 a scheme of the Wireless Sensor Network to monitor the Agrisfera irrigated area is shown.

Sub-action B.5.1 Installation of the hydro meteorological station (month: from 11 to month 22)

Sensors in irrigation channels - A sensor to measure the water flow and its electrical conductivity will be installed at the entrance of the irrigation channels in the study area. These sensors will be enough to know the quality and flow of the irrigation water, since no other water supply is used in the irrigated area.

Sensors in drainage channels - A complex network of drainage channels goes through the irrigated area. One sensor in each main drainage channel will be enough to know the salinity and flow of drainage water out of the irrigation district. The location of these sensors will be selected on basis the study performed in Action B1.

At this stage, it is envisaged that a total of 13 sensors will be enough for monitoring the irrigation and drainage water level and salinity. Each sensor will send the information wireless to a gateway which will resend the data by GPRS/CMDA to the central control.

Installation of soil sensors and piezometers - Initial estimates envisage that a total of 12 soil sensors will be enough for monitoring the area in a first approach, but this number could be modified depending on the results of the soil spatial analysis performed in Action B1.
The soil sensors will be installed at 40 cm depth for measurement of the soil water content and electrical conductivity at 25 °C. The specific sites for the installation will be selected on basis the spatial analyses carried out in Action B1, to monitor the most representative fields in terms of crop and soil. The depth of 40 cm will be selected because it is the bottom of the rooting zone where most of the water and nutrient uptake

is carried out. Each sensor will be linked with a central node or gateway via wireless signal. Each node will receive data from the soil sensor and will send it to a proximal repeater (or gateway) located at a distance less than 750 m, which is available via direct wireless communication. The sensor control unit is also inserted in the soil using an auger to reach the installation depth, so that only the antenna cable is left above ground. The antenna is positioned so that it is free from disturbance of the plant canopies. Ideally 0.5 m above the crop is enough so as to minimise wireless signal interference and also reflections.

Initial estimates envisage that a total of 12 sensors will be installed for piezometric level monitoring. Each piezometer will be installed close the field where the soil sensors are installed and in close enough distance to be able to transmit data towards the gateway or another node of the heterogeneous WSN that acts as a repeater.

The gateways will be located with respect to distance available for direct wireless communication to receive the signal from the repeaters and retransmit the data by GPRS/CMDA to the central control server (Figure B5.1). This Wireless Sensor Network uses robust and reliable operational architecture based on the WaterBee and SMART-PADDY systems.

Weather station - A complete weather station (Figure B5.2) will be installed in a secure place of the Agrisfera area. This weather station will provide data for the local irrigation management. The technical components of this station will be:

- Sensor for temperature
- Sensor for solar radiation
- Anemometer
- Pluviometer
- Sensor for atmospheric relative humidity
- Datalogger
- Outdoor box
- GPRS/GSM connexion.

Sub-action B.5.2. – Installation of gateways and connection to server (from month 12 to month 23)

Gateways and connection to server - Four gateways will be installed by initial estimates (Figure B5.3). The gateway will have the function to receive all the data from the sensors which will be sent by the repeaters and retransmit by GPRS or CDMA to the central control server. This server will be located in a secure place of the Agrisfera area at less than 5 km distance from the furthest gateway.

The Gateway Unit will be adapted from the ones used in the WATERBEE and SMART-PADDY projects (Figure B5.3) within Action B4.4 in order to integrate the heterogeneous sensor nodes: piezometers for water-table

measurement, soil sensors for electrical conductivity measurement, water content and temperature (sensor GS3, Decagon) and water salinity sensors in the drainage and irrigation channels.

Expected results:

- The main result of this action is the Wireless Sensor Network installed in field and operating correctly: taking data and sending them to the remote control centre.

- By this action the sectors of the pilot area will be monitored in terms of climate, soil salinity, soil water content, irrigation and drainage water salinity and volume, and water-table depth.

Indicators of progress:

- Number of soil sensors installed and operating.

- Number of water sensors installed and operating.

- Number of piezometer sensors installed and operating.

Beneficiary responsible for implementation:

OSV

All partners are involved in the Field installation in an equal way in supporting OSV.

Assumptions related to major costs of the action:

Travels (17.320 €)

Being the action in the pilot field, all the partners will be equally involved in travel (about 5.600 € per partner).

Equipment (3.500 €)

3.500 € , with eligible cost after depreciation of 1.400€ : Weather station

Consumables (15.500 €)

4.500 €: 12 soil sensors for umidity, salinity and temperature, with components and connections for Italy

7.000 € : 20 level, temperature and salinity sensors with piezometer and accessoires for Italy

4.000 € : Solar powered supply and additional rechargeable battery system for all the sensor node and gateway for Italy

Name of the picture:

Figure B.5.3 – Waterbee and Smart-Paddy Gateway node and its installation in field

Figure B.5.2 – Weather station

B5's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
Short report about the correct operation of wireless sensors network	09/2018

B5's PROJECT MILESTONES

Milestone name	Deadline
Wireless sensors network and weather station fully installed	07/2018
First set of nodes and gateways for the preliminary functioning of the WSN	05/2017

B. Implementation actions (obligatory)

ACTION B.6: Field application of SMART-AGROWETLANDS II and evaluation of its effects on crops

Description and methods employed (what, how, where, when and why):

The action B.6 has two sub-actions.

The first one, B.6.1-Field application of SMART AGROWETLANDS II aims

to verify how the model of SMART-AGROWETLANDS II system works in the field when the farmer is free not to use the system suggestions (system tested in open loop functioning)

to tune it finally so that it responds to crops requirements reducing excessive water level and salinity excursion (closed loop functioning of the tuned control system aiming to reduce crop stress).

In the absence of this tuning, even a potentially very good system could produce unsatisfactory results.

The second one, B.6.2-Evaluation of the effects of SMART_AGROWETLANDS II on the crops, aims to evaluate how actually SMART_AGROWETLANDS II works in its final form, evaluating the effects of the new water management on the crops and their productivity. This activity is necessary to give a more concrete evaluation of the effectiveness of the system. This aspect is important also in anticipation of the activities of dissemination, training, replicability of the actions D1 and D2

Every third month, an evaluation of the system will be performed to ensure its correct operating conditions. Several changes in the initial WSN-Web system configuration could be carried out to adapt the system to particular conditions, critical values etc. within the pilot area.

Sub-action B.6.1 - Field application of SMART AGROWETLANDS II – This sub-action will start on May 2017 and will last on April 2018.

At the start of the B.6.1 action, the action B1 is going to be concluded and the new model of soil and water salinity for irrigation recommendation (action B.2) has achieved a significant progress. The same is for the actions B.3 and B.4. About action B.5, a first installation of the WSN in the pilot area has been just done (May 2017) . Therefore, the main conditions to start a preliminary field evaluation of the SMART-AGROWETLANDS II are satisfied. During the sub.action B.6.1 the system will work in “open loop” mode and the farmers will adopt the conventional management of the irrigation water and actual irrigation will be assessed every week.

The real system behavior will be recorded for at least the following parameters measured by the system itself:

- meteorological conditions,
- water content and salinity of soil,
- water flows and salinity in channels,
- temporal evolution of water-table level in each field.

The last three items will be compared with the value provided by the model. Results will be used to improve the model and verify its performance for the surveyed area.

Expected results

- Errors and failures within the system are fixed.
- Improvement of the model and of the DSS of SMART-AGROWETLANDS II
- First demonstration of the suitability of the system for the integrated water management in the pilot area.

Indicators of progress:

- Temporal evolution of soil salinity average values in the fields, represented by the sensor measurements of soil electrical conductivity.
- Temporal evolution of water flows and salinity in channels.
- Temporal evolution of water-table level in each field.

Sub-action B.6.2-Evaluation of the effects of SMART_AGROWETLANDS II on the crops.

This sub-action will start on March 2019 and will last until February 2020. At the start of this sub-action the field application of SMART-AGROWETLANDS II system, in “open loop” mode, has just been completed. The system is now ready to work in “closed loop” mode and the farmers will adopt the irrigation recommendations of the system itself.

Every second week the real system (crops) and the WSN (sensors) will be checked.

In order to have an evaluation of the crop water stress, field measurements of the NDVI (Normalized Difference Vegetative Index, Raun et al., 2002-Agronomy Journal 94: 815-820) parameter will be made by handheld crop sensors (Green Seekers) on six georeferenced plots, 8m^2 each, surrounding each soil sensor, at different distance and direction from the sensor itself.

Every second week during the dry season, a drone equipped with a thermographic camera (infrared camera) will take a thermography of some (3-5) experimental fields, managed by SMART-AGROWETLANDS II. Areas with different temperatures (different colors) will show different water availability for the plants and water stress situation, when temperature is significantly higher than a calibrated thermal threshold.

Finally, in order to have a systemic description of the entire pilot area under the effects of the water management by SMART-AGROWETLANDS II system, as well as under the conventional irrigation management, one aerial flight will be done to take thermography with high resolution, showing water stress situation over all the pilot area at a given day.
This flight is foreseen at the start of June 2019, when winter and summer crops are both present in field, even if in different phase of their cycle.

At the crop harvest (end of June for winter crops, mid September for summer crops) the effectiveness of the irrigation strategy will be evaluated by the obtained yield (q/ha) in areas managed by SMART AGROWETLAND II, compared with the yield of areas managed by conventional irrigation system. If it will be possible, the yield of the georeferenced plots (8 m^2) where the NDVI was periodically measured during the growth season, will be measured separately.

All the data gathered during the sub-action B.6.2 will be analyzed, statistically processed and organized in a final report on the SMART AGROWETLANDS II performances, to be produced within January 2020. Some of the technical contents of this report will be used, in popular form, for the actions of dissemination, networking, training and replicability (D1 and D2) foreseen at the end of the project.

The final report will allow to evaluate the performance of the SMART-AGROWETLANDS II system. Good performances before tuning will represent a very positive element for the replicability of the system, even in situations different from those of the pilot area. On the other hand, not satisfactory responses will provide useful elements to a better tuning the system, assessing whether the unsatisfactory answers are due to deficiencies intrinsic to SMART-AGROWETLANDS II or to the modeling or to an insufficient environmental characterization of the area.

Expected results

- Demonstration of the suitability of the system for the integrated water management in the pilot area

- Water irrigation reduction by around 15% and soil salinity reduction by around 10% in comparison with the initial conditions (Action B1).

- Productivity increase by around 10%

Constraints and assumptions:

The irrigation recommendations must be followed by the farmers.

The salinity of irrigation water should not change much as this would invalidate the irrigation recommendation

Beneficiary responsible for implementation:

UNIBO

UNIBO will be assisted by the Agrisfera co-operative farm in the field activities, by OSV for technical problems

Assumptions related to major costs of the action:

Travels (3.488 €)

All travels are concerning UNIBO staff (being Agrisfera already *in situ* and not being needed for OSV) Travels have been reduced as much as possible but are necessary for the high number of activities in the pilot area and surveys foreseen in this action.

External Expertise (20.000€)

Rent of airplane: 1 flight with thermography and complete survey of the pilot area (10.000€) (Agrisfera)

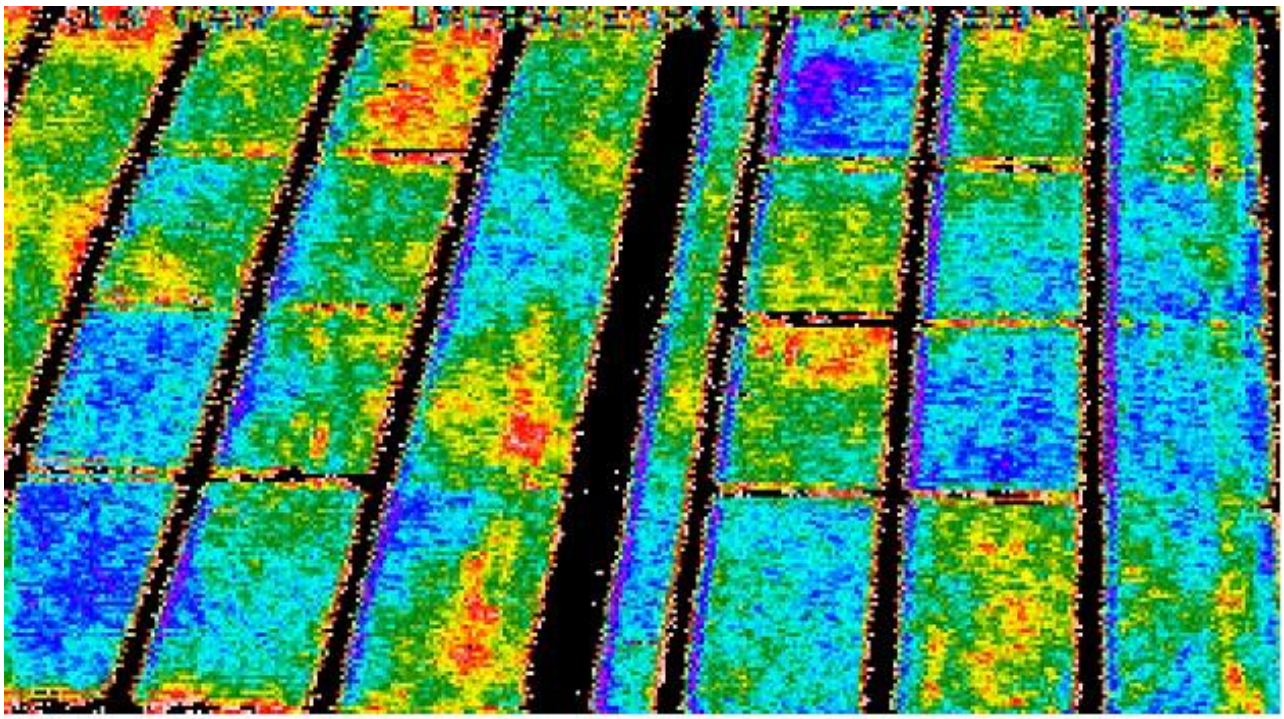
flights of rented Drones for field monitoring of water stress by thermography, 10 flights x 1000 eur (UNIBO)

Aerial thermal surveys require instrumentation not available to partners (rental=most economical solution). The use of both is related to different scale and time repetition.Thermal flight (just once)covers an area much wider than drones.Drone survey is carried out several times during the crop growing season to document the distribution over the fields of crop characteristics to compensate for the border effect of measurement taken by the "greenseeker"

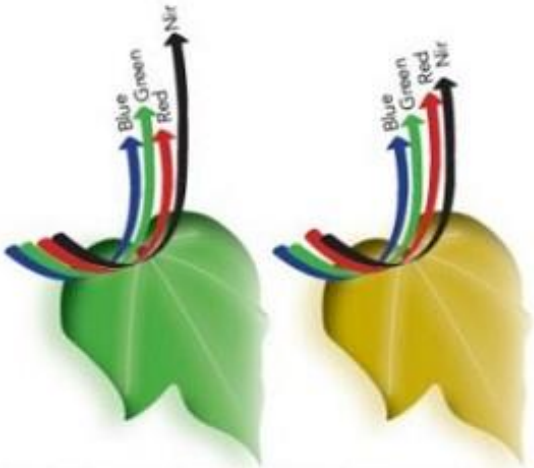
Equipment (1.800€)

1.800 total cost (900€ only as eligible) for 3 green seekers (600€ x 3) (used in both C1/C2 and B1/B6 but mostly in C1)

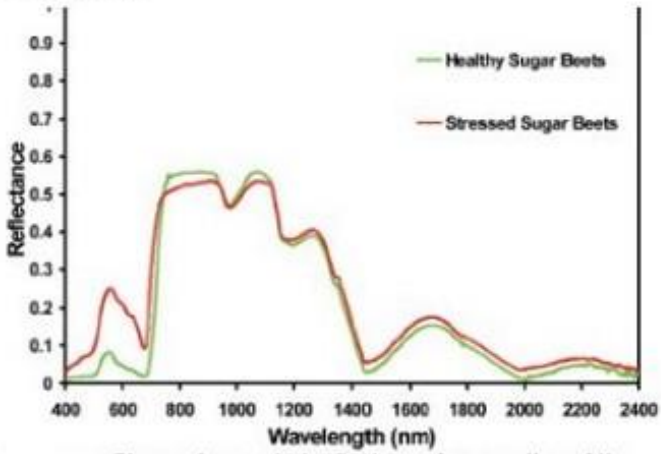
Name of the picture: Figure B6.1 Thermal image of a cotton canopy that was part of a water and nitrogen study in Arizona and Differences in reflected light between an healthy and unhealthy leaf



Thermal image of a cotton canopy that was part of a water and nitrogen study in Arizona. Blues and greens represent lower temperatures than yellow and orange. The image was acquired with a thermal scanner on board a helicopter.
From: http://www.uswcl.ars.ag.gov/epd/remsen/irrweb/thindex.htm



Differences in reflected light between a healthy and unhealthy leaf.



From: www.aces.edu

Change in spectral reflectance for a portion of the EM spectrum for a healthy sugar beet plant and one under water stress.

B6's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
SMART AGROWETLANDS II system report: closed loop behavior, assessment of performance	01/2020

B6's PROJECT MILESTONES

Milestone name	Deadline
Start of the AGROWETLAND system operation	07/2018
Full AGROWETLANDS system operating in the pilot area (after adaptation and validation)	09/2019

C. Monitoring of the impact of the project actions (obligatory)

ACTION C.1: Monitoring the Project's impact on salinity status in agro-wetland pilot area

Description and methods employed (what, how, where, when and why):

Sub-action C.1.1 Reports on the outputs and impact of the project (Life project performance indicators)

The action C1 will start on October 2017 and will last to the end of the project. The early start of the action wil be only dedicated to the report on the outputs and impact of the project taking into account the LIFE project performance indicators, at the time of the 4 foreseen mandatory activity reports.

Sub-action C.1.2 Monitoring the Project's impact on salinity status in agro-wetland pilot area

The Sub-action C1.2 will start on September 2019 and will last to the end of the project. The soil sampling in 50 plots of the Agrisfera pilot area, will be carried out at the beginning of October 2019, at the conclusion of the agricultural season 2019.

The main objective of this action is to evaluate the impact of the project actions in terms of salinity status of soils, drainage water flows, as well as volumes of water used for irrigation of the area. This action will be focused on the comparison between the scenario described within Action B1 "Environmental database construction, modelling of water and salt flow in the pilot area" and the scenario after the implementation of the SMART-AGROWETLANDS II system, as well as the separation of differences observed in the field between the systematic effect of the system and the effect of uncontrolled vagaries of weather and hydrologic conditions.

A second measurement of water quality after 1 year at least of irrigation management carried out with the aid of SMART-AGROWETLANDS II system is necessary to evaluate effects of the new strategy and to obtain some information about system uncontrolled variables and modeling errors.

For this reason the monitoring action C1 will play a particular role in monitoring the impact of the project.

Three tools are available for this comparison:

12 time histories of water flow and elevation in the area, salinity in soil and channel water, one meteohydrological record, recorded by the SMART-AGROWETLANDS II monitoring system;

50 plots where soil and water salinity can be compared at 2.5 years lag;

a model of the hydrologic behaviour of the area in the real period as well as in several other representative meteo-hydrological conditions in the same season.

The soil salinity will be monitored using the web SMART-AGROWETLANDS II system in the Agrisfera pilot area. On the basis of the measurements performed in the 12 instrumented areas during the 1 year at least of the

system application (sub-action B.6.2), we will obtain a first point estimate of the temporal evolution of soil salinity.

In order to know the status on the whole pilot area, a new field survey, visiting the same 50 points as in the Action B1, will be carried out in the last year of the project. Only the soil water salinity will be determined using the same laboratory protocols since grain size composition and physical properties of the soil are nearly invariable over such short time interval. Punctual comparison of soil conditions in the two scenarios will be performed on the whole set of the 50 field sampling points. Integrating the field data with the web data, the impact of the project implementation on the pilot area will be determined.

The comparison of the real behaviour and model simulation for the specific monitoring period with simulation of weather conditions will show what can be the SMART-AGROWETLANDS II systemic effect in comparison with natural variability.

Since the irrigation, the drainage flow and salinity are monitored by the web SMART-AGROWETLANDS II system, this allow us to estimate the salt input and output in the pilot area, and a water and salt balance of the area in the monitoring period can be performed showing irrigation strategy effect at an aggregated scale rather than punctual.

The information will be used to evaluate the salt load to the wetlands and the salt input in the cultivated fields. Effect of evaporation in concentrating salt in the irrigated areas will be assessed, that will be considered to evaluate the water use efficiency in terms of salt balance. Through the monitoring of soil and water salinity, the recommendation to farmers for water irrigation management can be opportunely modified to optimize the water use in terms of efficiency, soil salinity, water salinity or flow. These recommendations will be generated by the SMART-AGROWETLANDS II system.

Expected results:

Statistical information about system uncontrolled variables and modelling errors will be obtained.

Regarding the soil salinity within the agricultural fields, we expect a reduction that keeps it below the harmful levels for the medium sensitive crops of the area, such as wheat or sorghum. For the most sensitive crops, such as alfalfa (a marginal crop in the pilot area and in its surroundings) with a threshold ECe equal to 3.4 dS/m, the harmful level could sometime be exceeded.

We expect also a 10% decrease in the amount of salts exported from agricultural fields to the natural wetlands.

A Lyfe Cicle Assessment will be produced at the end of the project, as an annex, following the Final Report of evaluation of water use efficiency, soil salinity and salts load to the wetland within the pilot area.

Beneficiary responsible for implementation:

UNIBO

UNIBO will be assisted by Agrisfera co-operative farm in the field activities

Assumptions related to major costs of the action:

Travels (3.120€)

Travels have been reduced as much as possible, but are necessary for in field monitoring from the staff.

Consumables (4.400€)

500€: Plastic bags, plastic pots and other field material for taking samples.

1.800 € : Chemical reagents for water laboratory analysis of 300 samples= 300*6 €

1.600 € : Chemical reagents for soil laboratory analysis to determine of soil organic matter, carbonates, ph and electrical conductivity, and water salinity for 200 soil sampling at 8 €/ sample

C1's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
updated LIFE Project Specific Indicators Call excel table	12/2017
updated LIFE Project Specific Indicators Call excel table	06/2018
updated LIFE Project Specific Indicators Call excel table	06/2019
final LIFE Project Specific Indicators Call excel table	09/2020
Complete life-cycle assessment on positive and negative environmental impacts of LIFE AGROWETLANDS II system for water and soil salinity management	02/2020
Final Report of evaluation of water use efficiency, soil salinity and salts load to the wetland within the pilot area	02/2020

C1's PROJECT MILESTONES

Milestone name	Deadline

C. Monitoring of the impact of the project actions (obligatory)

ACTION C.2: Monitoring the Project's impact on the natural and semi-natural habitats, plant communities and biological quality of water and assessment on socioeconomic impact

Description and methods employed (what, how, where, when and why):

The action C2 will start at January 2017 and will last to the end of the project (February 2020). Beside salinity, the project will significantly impact on natural and semi-natural environments and on socio-economy. Relations between the two sectors are week and can be naturally subdivided in two sub-actions, each one dedicated to one sector.

Subaction C.2.1 – Monitoring project impact on the natural ad semi-natural habitats, plant communities and water quality

The aim this action is to check the effects of the proposed management on the environmental quality of natural and semi-natural habitats, as well as on the species inhabiting them. In particular, the five farms of Agrisfera, where the experimental activities of this LIFE project will be carried out, are quite close to a significant number of SCI/SPA of the European Ecological Network Natura 2000. This is the reason because a specific monitoring activity of the project's impact on the natural and semi-natural heritage of the territory, where the project itself will take place, is appropriate.

From West to East (Figure C2.1), the protected areas in question, are: the northern part of the SCI/SPA IT4070001 Punte Alberete, Valle Mandriole, the SCI/SPA IT4070002 Bardello (99 ha), the Northern part of the SCI/SPA IT4070003 Pineta di San Vitale, Bassa del Pirottolo, the Northern part of the SCI IT4070005 Pineta di Casalborsetti, Pineta Staggioni, Dune di Porto Corsini. South of the Lamone River, a wide area, resulting from the Southern portions of the SCI/SPAs already mentioned, as well as by the SCI/SPA IT4070004 Pialasse Baiona, Risega and Pontazzo of 1596 ha, is present. They include habitats, flora and fauna species (more than thirty different bird species and many invertebrates) of community interest (Habitat Directive 92/43/EEC) some of them of priority importance, such as the plant species **Salicornia veneta*, the amphibian species **Pelobates fuscus insubricus* and the habitats 2130 - * Fixed coastal dunes with herbaceous vegetation ("grey dunes"), 2270 - * Wooded dunes with *Pinus pinea* and/or *Pinus pinaster* , 3170 * Mediterranean temporary ponds. The SCI/SPA Bardello hosts the largest brackish grassland (one hundred hectares) of the North-Eastern Adriatic coast. Coastal erosion, subsidence and ingression of saline wedge and sea water are among the main factors that threaten these areas and their habitats and species.

Another interesting area, from the naturalistic point of view, is the Chiavichino farm, located to the North of the SCI/SPA Bardello. It represents a semi-natural area, since it was withdrawn from production (expiry 2020) and reforested with native hardwood species, in application of the EEC Regulation 2078/92 and the Regional Plan of Rural Development (EC Regulation 1257/99) for the re-naturalization of the agricultural lands.

The five sites indicated in figure C2.1 will be selected as monitoring sites of the project' impact on the natural and semi-natural environment. Field surveys and description on habitat, plant communities and saprophytic fungal communities, biological quality of water will be performed in the period January 2017-September 2017, at the beginning of the project, in almost 30 georeferenced plots (25 for terrestrial habitats, each of about 100m^2, 10 for aquatic habitats). Biological quality of water will be evaluated by an Index of Biological Integrity (IBI), based on the specific biodiversity of the benthos invertebrates and on the presence and abundance of halo-tolerant species. The samples of benthos will be taken in June-July 2017, the optimal season for an exhaustive analysis of these communities. Biological quality of the terrestrial communities will be evaluated through indexes of species diversity.

The same set of field surveys will be repeated during the same months of 2019. A report will be produced on the results of the two field surveys and on the emerging trends after almost one year of the application of the SMART-AGROWETLANDS II.

Constraints and assumptions

- The physico-chemical analysis of discrete water samples do not tell the whole story about the conservation status and environmental quality of the sites, because they represent just some points samples from a strongly variable history. Conversely, the residential flora and fauna integrate the effects of environmental conditions and, therefore, an index of biological integrity (IBI) better reflects the environmental quality of the monitored sites.

- Since the access to the protected areas for samplings could be limited, an access authorisation will be previously requested to the competent authorities.

- Possible delay of about 2-3 months in the authorization process for access to the protected areas and sampling: to avoid the risk, UNIBO will request immediately the authorizations, during Summer 2016, before the project beginning. In case of delay, the 6+3 month buffer added to the whole project for administrative problems and project's closure should cover the risk. No budget increase should be necessary for this particular risk.

Espected results

Similarly to the previous C1 Action, no detailed quantitative information about the environmental quality of the monitoring sites is available. Therefore we specify the expected results in relative terms, with reference to the status at the beginning of the project. We expect that the environmental quality will increase, we expect at least 10% decrement in nutrients (nitrogen, phosphorus) and salt loads in waters. As a consequence the Index of Biological Integrity (IBI) based on the benthos invertebrate community will experience 10% increase relative to the first assessment.

Sub-action C.2.2 Assessment on socio-economic impact

An assessment on socio-economic impact will be at disposal under the form of a report. A poll will be passed to the farmers in the fourth quarter of 2019. At this date the first results of the comparison between yields obtained at Agrisfera farms, with or without the irrigation managed by SMART AGROWETLAND II, will be available. All the quantitative information of the system performances will be presented. The farmers will be invited to express their opinions, deriving from their personal experience, their perception of the problems as well as from the situations they usually manage.

3 sections :

- Water management questions: to find out how farmers perceive the improvement in the integrated water management.
- Environmental questions: to know the social perception of improvements in the wetland habitat and soil salinity.
- Economical questions: to know the improvement in the marketable yield and the economic profit.

The study will show the impact on supplementary income sources, effect of the raised profile of the area, naturalistic tourism, quality of the food production, linked market.

Expected results

The farmer perception of water management efficiency and soil - water quality is expected to improve, and this will be shown by the results of the poll.

Beneficiary responsible for implementation:

UNIBO

Agrisfera will support UNIBO in the actions in the field

Assumptions related to major costs of the action:

Travels (1.560€)

Travels have been reduced as much as possible, but are necessary for in field monitoring

External Assistance (*5.000€*)

5.000 for a Zoologist for animal species identification (water quality bio-indicator)

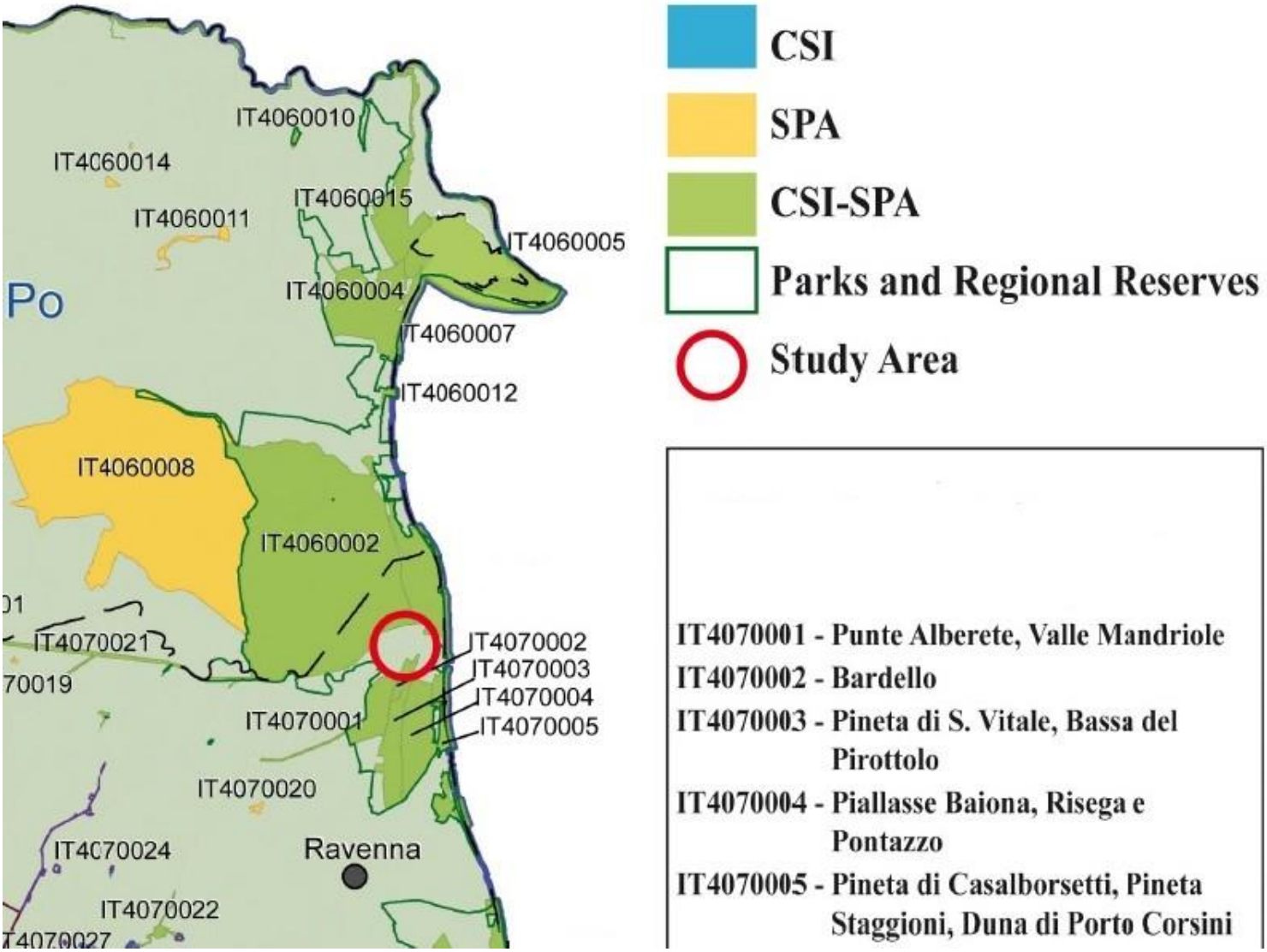
Equipment (1.800€)

1.800 € (only 900€ as eligible) for a field laptop for elaboration data analysis for 33 months on 36 total duration of the project (used also in C1)

Consumables (2.175€)
Chemical reagents for water laboratory analysis of 200 samples during the different sampling events at 6 €/sample

Biological laboratoy analysis to determine of 20 bhentos and water samples from the ponds =20*40€/sample

Name of the picture: Figure C2.1 - Five natural and semi-natural protected areas



C2's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
Report on the improvement of the water quality parameters along the development of the project and its relationship with the irrigation scheduling followed in the Agrisfera's district	03/2020
Report with the main results f the socio-economic poll to farmers	12/2019
Report: Assessment on socio-economic impact	02/2020
Guide for recommended measures aimed at the improvement of water quality and quantity of freshwater ponds in similar agro-wetlands	03/2020

C2's PROJECT MILESTONES

Milestone name	Deadline

D. Public awareness and dissemination of results (obligatory)

ACTION D.1: Communication Pack and Dissemination Activities

Description and methods employed (what, how, where, when and why):

This action aims to produce tools and relevant material to be used for the dissemination activities of the project. It includes both the mandatory deliverables (website, notice boards, Layman's report) and the dissemination material aimed at the stakeholders and general public. A part of the material produced will be also used in the action D2, focused on replicability and training actions.

A series of simple, small and efficient activities will be undertaken for the dissemination fo project reults to the stakeholders and general public.

- regional wetland growers (> 100 regional growers will be targeted)
- other cultivators (> 500 will be targeted from wetland regions across Europe)
- other industry stakeholders (> 500 will be targeted from relevant sectors from the food supply chain and water management and environmental sectors at European level)
- policy makers (> 100 at European level)
- standardisation and environmental bodies (> 50 at European level)
- consumers and general public (> 3000 through press releases and other actions)
- research community (> 500 at European level), etc.

The stakeholders at both national and European level will also be targeted in the training activities developed in Action D.2. Moreover, stakeholders beyond those at regional level will be approached to trial the outputs of B3.

The coordinating beneficiary and associated beneficiaries will acknowledge the support given by the Union in all documents and media produced in the framework of the project, using the LIFE logo provided by the Commission.

Sub-action D1.1 Development of the Dissemination Pack's mandatory deliverables: website, notice boards, Layman's Report).

Website

A project website will be constructed and actively maintained as a powerful major dissemination tool (it will also house a restricted access area for the partners for storing information and facilitating internal communication). The public site will present the project brand, project information, its partners, its core objectives, the environmental and economical benefits associated with the SMART-AGROWETLAND technology, the LIFE logo and information about the LIFE programme, etc. The website will report the objectives, actions, progress and results of the project. Additionally it will contain a Newsroom & Events section and an on-line training section. The website will be up already at the beginning of the Project and will be continuously updated. At the end of the project, the relevant public information and results will be

available at this site that will be maintained online for further 5 years. The project leaflet, posters and other promotional materials will be also included.

The site has to present important messages to the external public; a potential problem is to find the correct setting for the message to reach the target. For this reason this work requires an intervention of a specialized personnel. For this reason to build this part the project needs a sub-contractor. The remaining risk of minor or insufficient updates after the project’s end will be countered by the After-Life Plan.

Notice boards

The coordinating beneficiary and associated beneficiaries will erect and maintain LIFE information boards describing the project at the locations where it is implemented and in particular in several strategic zones of Agrisfera area and in Spain during B6 implementation.

Layman’s report

The project will produce the Layman's report in the last quarter of the project implementation. The Layman's report of 5-10 pages will outline the main results of the project targeting at a non-specialist audience. It will be available in paper (500 copies) and electronic format for download at the project website. It will include comprehensive results from the environmental impact of the soil monitoring system in the installation in Agrisfera agro-wetland.

Sub-action D1.2 Production of communication/dissemination materials and Communication Plan

A Communication Plan will be established to reach both stakeholders and the large public. At the same time, to reduce the carbon footprint, the use of paper has to be reduced, thus the priority will be given to digital tools. The Communication Plan will take the form of a guideline and foresee actions and tools aiming to publicize the project and its results and to support the communication, awareness and training actions so to raise target stakeholders’ awareness on demonstrating the problem targeted.

Beside the obligatory dissemination activities (website, notice boards, Layman's report), the Communication Plan will include the use of:

Project brochure: both digital and on paper. The paper ones will be reduced at only 1000 copies to be used during workshops. Target: stakeholders.

Project posters: posters (10 copies) with the LIFE logo will be prepared and exposed at the final conferences and workshops (sub-action D.2.3.) and during networking activity events with other projects (sub-action D.2.1). Target: stakeholders.

Press Office: it will organize press conferences, write press releases and newsletters, and make the press reviews on local and national printed and web thematic and general press and media. It will collect all the news about the local area in thematic press reviews. The press releases will be sent to specialized Italian and international magazines, and the newsletters will be published regularly on the web site of the project. The press review will be realized and organized for typology (i.e. video, web, printed media) by scanning the local printed press and recording the videos and will follow all the project. Target: stakeholders, citizens and large public.

Expected results and outputs

- 1 Communication Plan

- 1500 project brochures in English(500), Italian (500) and Spanish (500);

- 1 project poster in Italian and Spanish (10 copies on each language);

- 3 press releases (kick off, action BXX results, project's end)

- 2 press conferences at the beginning and at the end of the project, together with kick off and final conference for XX stakeholders and journalists

- 1 press review for each year of the project (digital);

- 1 newsletter every 6 months (about 6 newsletter published on the website and sent to the stakeholders).

Beneficiary responsible for implementation:

UNIBO

IRIS, OSV and Agrisfera will give the necessary support to the communication manager for the Dissemination pack realization.

Assumptions related to major costs of the action:

All partners will be involved.

External Assistance (6.500€)

3.000 € for a website and assistance in regular updating normal cost for brochures,

posters, information boards and the other dissemination material

D1's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
Press release n°1 Agrowetlands Kick-Off	11/2016
Press release n° 2 - on the Start of the AGROWETLAND system operation	07/2018
Press release n° 3 - Agrowetlands Final Conference announcement and results	01/2020
Project's Brochures and Posters	09/2017
Agrowetland Communication Plan	11/2016
Information Board	09/2017
Project Website	11/2016
Layman's report	03/2020

D1's PROJECT MILESTONES

Milestone name	Deadline
Project's website online	11/2016
Final Conference of the Project	02/2020

D. Public awareness and dissemination of results (obligatory)

ACTION D.2: Networking, Training activities and Dissemination Events - Replicability

Description and methods employed (what, how, where, when and why):

This action will start soon after the project's beginning with the networking sub-action. Togheter with the networking, the training activities represent an essential tool to transfer and replicate the knowledge on the DSS and WSN systems and its benefits to the end-users. At the end of the project the training will be done in 5 special session as open training for target groups. A small replication activity will be held in Spain.

Sub-action D2.1. Networking with other projects

The experience of past and on-going LIFE projects dealing with the same or similar problems will be important for both increasing the range of the effect of the project’s outputs (replicability) and sharing experience.

Networking activities will aim to exchange knowledge and best practices among similar project platforms to ensure an efficient transfer of know-how and experience in order to foster its replication in similar context.

Some LIFE Projects subject to being contacted are:

- LIFE SEGURA RIVERLINK – RIVERLINK. LIFE12 ENV/ES/ 001140.

- IRRIGESTLIFE- TELEMANAGEMENT NETWORK USING FREE CONTROLLERS CONECTED TO A GIS FOR AN OPTIMIZED IRRIGATION IN VITORIA-GASTEIZ LIFE11 ENV/ES/000615.

- HUMEDALES DE LA MANCHA - Restoration of salt flats around 27 endorheic wetland areas in La Mancha LIFE10 NAT/ES/000563.

- Δ-LAGOON - Restauración y gestión del hábitat en dos lagunas costeras del Delta del Ebro: Alfacada y Tancada LIFE09 NAT/ES/000520

- IES - Irrigation expert Simulator. LIFE11 ENV/ES/ 000621

- WSTORE2 – Reconciling agriculture with environment through a new water governance in coastal and saline areas(LIFE 11ENV/IT/035).

Once a number of these LIFE projects have been selected and contacted, agreements will be made with project managers to hold a meeting to find possible synergies and collaborations between projects. These collaborations could be focused on.

- Identifying resources for the action of networking and dissemination (link on their respective websites, etc)

- Verification of possible future interactions between the actors involved

- Exchange of information material for mutual support in the actions of dissemination and networking.

Sub-action D2.2. Training programme and materials, publications

Publications:

A final publication (booklet), plus Scientific publications (2 for engineering aspects and 2 for agronomic) will be produced.

Training:

These sessions will be prepared by UNIBO and OSV and delivered to the farmers of the Agrisfera agrowetland and the interested stakeholders or member of the target groups.

The aim of these sessions will be to educate participants on the technical use and operation of the SMARTAGROWETLAND environmental instrument at various stages of the field validation process, as well as to explain the competitive and environmental benefits of the uptake and use of the different results of the project. This will facilitate understanding, acceptance and future uptake and exploitation of the results of this project by farming communities.

During this action the programme of sessions will be drawn up and planned to coincide with the evolution of the work- dates will be finalised, the target group and attendance capacity defined, how attendees will be approached, how these sessions will be conducted, evaluated, etc. In addition the necessary materials that will be needed for each session will be planned to ensure that they are prepared and validated in time for each session. Furthermore, a feedback questionnaire will also be designed in order to capture any worthwhile comments, impressions, ideas, and improvements from the target audience.

Moreover, an on-line training tool will be embedded in the project website, and the materials will be available on-line for the general public to download and therefore, a wider audience of farmers across Europe, and other interested stakeholders (such as Agricultural associations, agricultural ministries and policy makers) who cannot physically attend the sessions will benefit (the on-line training tool will be disseminated to over 1000 farmers and stakeholders regionally, nationally and at European level).

In addition, the filming of one or more of these sessions will also be discussed and agreed with a view to heightening the impact of these sessions.

The training programme will be implemented in a timely fashion and all activities will be executed in keeping, as much as possible, with the above plan. At least **5 training sessions** with local farmers from the Irrigation Community of Agrisfera (> 60), other farmers beyond the Consortium (> 30), further stakeholders (> 10) and NGOs and environmental public bodies (>10). The 5 sessions will be used to explore all the main pillars of the AGROWETLANDS II method.

The training will be an important aspect of the project's closure, and will be connected and presented by the final conference organized together with the Emilia-Romagna Region (see letter of support), inviting stakeholders and sectors of the target groups to freely use the possibility to be trained in the system use.

Sub-action D2.3 Final Conference

One final conference will be organized by UNIBO in cooperation with the Emilia Romagna Region (letter of support), aiming at stakeholders at regional and national level (see the letter of supports of cooperatives, regional and national bodies annexed to the project). The participation to the conference will be connected

with the training activities forseen in Agrisfera, leaving the possibility for the attendants to subscribe for the training.

Sub-action D2.4 replicability in Spain

To test how the demonstrated features and techniques can be applied in other Mediterranean territories. This sub-action will last from April 2019 to February 2020 (11 months) and will be contemporary to the final part of the Action B6, when in the Italian pilot area the farmers will adopt the irrigation recommendations of the system itself.

It will be realized in the Spanish territory of the Comunidad de Carrizales (Elche-Alicante), a site of 1512 ha (76 % irrigated) used by 400 farmers. The irrigation water usually presents high salinity, which jointed to the clayey soil texture, the shallow piezometric level and the semiarid climate leads to an important risk of soil salinization.

A small area will be selected for the implementation of a little SMART AGROWETLANDS II system which will be used along all the 2019 growth season for one selected crop, in the “open loop mode”. This in order to tune the system to the requirements of the selected crop. IRIS will assist the Spanish responsible of the field experiments (indicated by the Comunidad de Carrizales), UNIBO and OVS will participate in the realization. All the farmers of the Communidad will be invited to follow the experimentation of the system.

First results of the Spanish experimentation will be presented at the Progress Meeting of the autumn 2019 (Alicante, Spain). The report will be finalized on Feb. 2020

Beneficiary responsible for implementation:

UNIBO

Agrisfera will host the training sessions, while IRIS will manage the project replication in Spain and will help UNIBO in Other LIFE projects networking

Assumptions related to major costs of the action:

Travels (10.695 €)

Travels are an important part of the action.

5 training session in Ravenna will see members of the Consortium partecipating as trainers or disseminators.

In the Networking with other LIFE projects at least 2 travels in Greece and Spain are foreseen for UNIBO and 1 for IRIS.

Travels aimed at starting the replicability actions in Alicante (Spain) will need travels from UNIBO and local partner (IRIS).

External Assistance (25.770€)

Around 10.000 € are foreseen for the 5 training session to be organized in Agrisfera.

A specific subcontract will be done by Agrisfera to an external network/cooperative for promotion towards Italian "Cooperative" in agricoltural sector Regione ER

The other external costs foreseen in action D2 are related to the catering and dissemination meetings organization:

-3.500 € final publication (UNIBO)

-1.500 € final conference organization and catering (UNIBO) -

400 € booklet for training in agrisfera (UNIBO)

Consumables (8.830 €)

The costs will concern UNIBO and IRIS for the replicability sub-action D2.4.

Other costs (4.320€)

Other costs are concerning the fees for the scientific papers publication (UNIBO)

IRIS Attendance to trade-shows in smart agriculture presenting the project in Spain (700€)

D2's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
Report on transferability of the AGROWETLANDS II system in the Comunidad de Carrizales	02/2020
Scientific Paper published (engineering aspect - project implementation)	12/2018
Scientific Paper published (agronomic aspect - project implementation)	12/2018
Scientific Paper published (agronomic aspect - project results)	12/2019
Scientific Paper published (engineering aspect - project results)	12/2019

D2's PROJECT MILESTONES

Milestone name	Deadline

E. Project management (obligatory)

ACTION E.1: Project management, monitoring and After Life Plan

Description and methods employed (what, how, where, when and why):

Agrowetlands is a transnational and public-private partnership in need of a sound management and monitoring. The management action will ensure the correct administration of the entire proposal and secure achievement of project objectives within budget, quality and deadlines. The Fig. E.1 resumes the rationale of the project consortium.

Sub action E.1.1 – Project management by UNIBO

The coordinator will elaborate and develop a **management plan** including guidelines for financial reporting and quality measures for deliverables and reports, to be presented to the partners at the kick-off-meeting (draft version) and in a final version 1 month after.

Implemented through the whole duration of the project (36 months plus 6 months of buffer), the action foresees a management structure explained in fig. E.2.

A **Steering Committee (SC)** led by UNIBO will include 1 technical representative for each partner (2 for UNIBO), the same people that will be individually charged of each action coordination. Each action responsible person collects the results of its action and transfers them as consolidated reports to the project manager. A general information flow is ensured during the meetings, where all partners inform each other regularly about their results. The SC is the decision-making body for daily matters regarding the execution of the project. The SC discusses and fine-tunes the basic work plan agreed with the EC. Any adaptation to the work plan is reported to the Consortium and to the EC. This process enables the SC to closely monitor the quality, tasks, budget and information within the project. The SC plays an essential role in the communication within the Project and from the Project towards third parties. Since action leaders are member of the SC, there is a direct interaction between it and action participants.

One full time manager will be employed, with experience in european project management and communication, and will be devoted to actions E1, D1 and D2.

A Consortium Agreement will be prepared as established in the Grant Agreement at the beginning of the project.

The first, mid-term and final reports will be produced, including external audit and financial statements.

Meetings

1 Kick-off meeting to be held in Bologna :

- management staff & management plan will be presented (Milestone E 1)
- review of the project with regard to time schedule and first actions launch

6 Progress meetings will be held in different partner’s location (3 in Bologna, 2 in Ravenna at Agrisfera and 1 in Alicante, Spain). Tasks:

- monitoring the fulfilment of the work programme, milestones, expected results, constraints and risk management
- preparing and approving the reports

1 Final meeting (Milestone E2, UNIBO): monitor the fulfilment of the whole work programme and the quality of the results obtained by the project during its duration, After Life Plan presentation, together with a final conference (see action D2.2)

Sub action E1.2 - Monitoring, Quality control and Risk Management

Monitoring, as a management tool for tracking progress of project, will be a continuous process throughout project implementation, gathering data about performance indicators and allowing a fast reaction in case of risks' manifestation.

Monitoring protocol

A **monitoring protocol** (UNIBO) will be ready under the form of a check list by using the indicators of progress defined by each action, deliverables and milestones. This check list will also indicate the action, the source of verification, the deadline, the name of the beneficiary responsible, costs, as well as comments if required.

The management monitoring, measuring the efforts in managing the project, will be connected to action C1 and C2, that will monitor at the opposite the results and impacts of the project.

The SC will asses an evaluation at mid-term (M21) and at the end (M42).

Audit report:

UNIBO will nominate the independent auditor that will verify the final statement of expenditure and provided to the EC in the final project report in respect with the Program General Conditions. The auditor will verify compliance with national legislation and accounting rules, and certify that all costs incurred comply with the grant agreement. The auditor will also check the sources of the project financing, and in particular that cofinancing does not stem from other Community financial instruments. The work to be performed will be in accordance with, and in the format of, the Guidelines provided by the Commission.

Sub action E1.3 - After LIFE PLAN

The After-LIFE Plan will be realized before the end of the project by means of the collected communication and dissemination best contributions. The Plan will put together all parts of the activities concerning Communication and Dissemination, aiming to the replication of the action, and include management of the realised plan in Agrisfera area and the replication activity in Spain.

The work to develop the After-Life Plan will begin after the first 3-4 months of the project, collecting (by all partners) all the usefull materials, tools, pictures and media. The Communication manager has to coordinate the activities. The guidelines of this Plan concern : 1) choice of the communication and dissemination channels, such as web sites, conferences and media, in parallel with 2) the choice of the main results and tools that need to be disseminated; 3) the preparation of the material that will be disseminated by means of each channel; 4) the implementation of each part of the After-Life plan for each prospective channel.

The After-Life plan will be available on the Project web site. In particular the Plan will remain on each Partners website for 5 years after the end of the Project. All the results and the tools of the project will have to be disseminated.

Beneficiary responsible for implementation:

UNIBO

Iris and Agrisfera, and in a minor part OSV, will contribute to the management, especially for the organization in Spain and on field in Italy.

Assumptions related to major costs of the action:

The other partners will provide the necessary working days to allow presence and coordination of their actions inside the managing structure of the project.

Travels (16.981€)

Travels have been reduced as much as possible, mostly related to the technical meetings and missions and actions on the field site from the locations of the partner's consortium. Travels in Spain for actions aiming to prepare the replication have been foreseen.

External Assistance (4.500€)

The project management will be carried internally. The only external costs foreseen are related to the catering and meetings organization.

Other costs (7.500€)

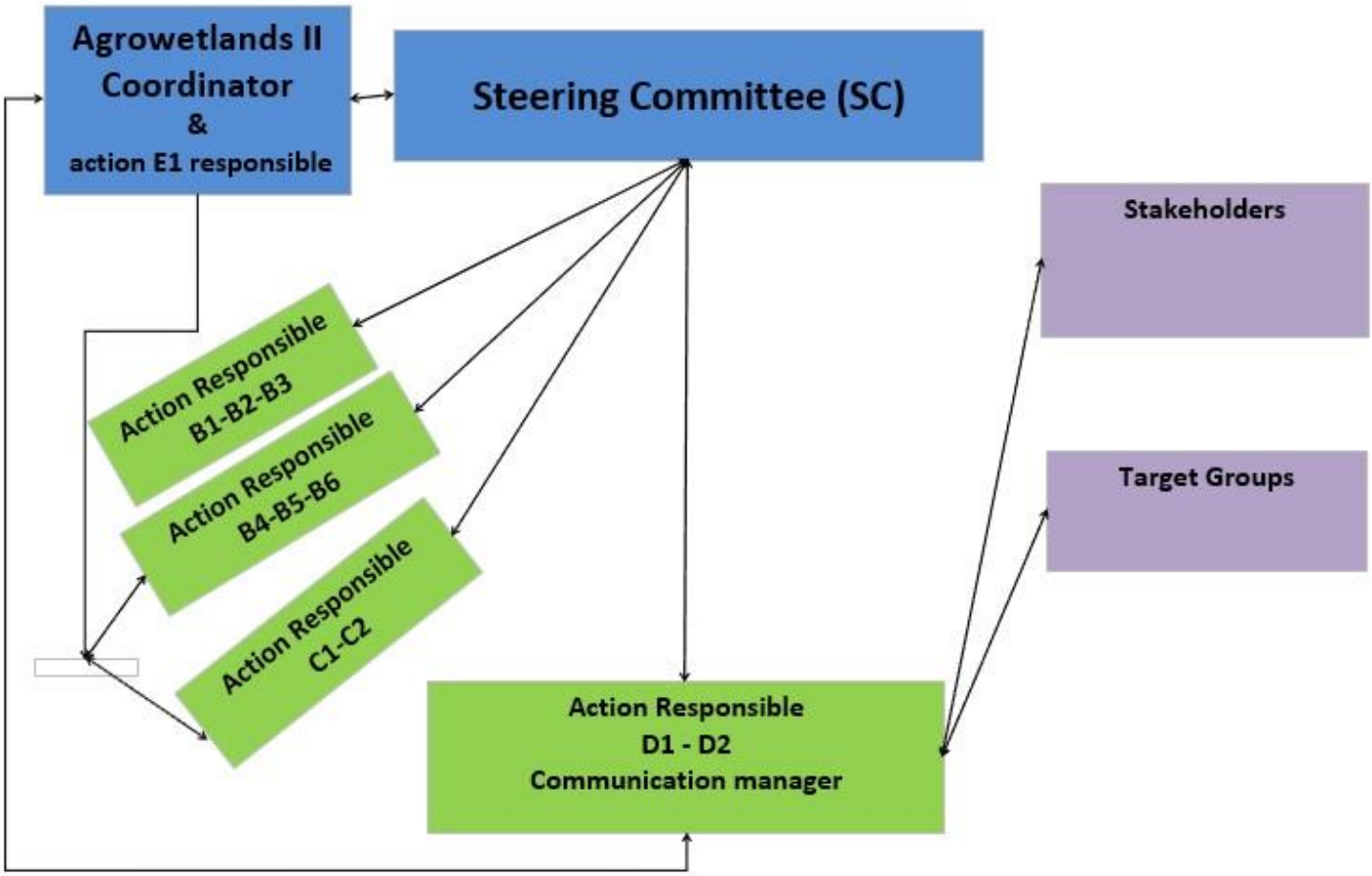
External Audit calculated on % on the project dimension (0,8%) for 7.500 €

Other costs are concerning the fees for the scientific papers publication

Name of the picture: Figure E1.1- Rationale of the Project Consortium

Rationale of the Project Consortium		
Beneficiary	**Country**	**Role in The project**
UNIBO	Italy	Responsible for Coordination and Monitoring, dissemination and communication including After LIFE Plan. Two different teams with different specialisation, agronomic (pedology and plant pathology included) and engineering (including hydraulics), will care about: the Environmental database construction, modelling of water and salt flow in the pilot area; development of the new model of soil and water salinity; will support IRIS in the development and implementation of web-control system and of WSN module to monitor soil and water salinity and water-table level for the sustainable irrigation; and will also coordinate Agrisfera and OSV in the Field application of WSN in pilot area. Responsible for actions B1, B2, B6, C1, C2, D1, D2, E1.
IRIS	Spain	Development and implementation of web-control system and user interface for irrigation management, development of Wireless Sensor Network (WSN) module to monitor soil and water salinity, and water-table level for the sustainable irrigation (from previous experience in Fp7 related projects); supporting OSV in Field installation of WSN. Responsible for actions B3 and B4
Agrisfera	Italy	Assisting UNIBO in Environmental database construction, modelling of water and salt flow in the pilot area; assisting and supporting UNIBO, IRIS and OSV for all actions taking place in the Pilot area; supporting Lead Partner in the coordination activities for the actions taking place in the pilot area. Supporting UNIBO in the dissemination activities aimed at regional SMEs and their associations and cooperatives in the same sectors.
OSV	Italy	Supporting UNIBO and IRIS for the development and implementation of web control system and WSN and user interface for irrigation management; responsible for the Field installation of wireless sensor network in pilot area (based on previous experience in related FP7 projects) . Responsible for action B5. Assisting Agrisfera and UNIBO in field application.

Name of the picture: Figure E1.2- Management Structure of LIFE AGROWETLANDS II



E1's PROJECT DELIVERABLE PRODUCTS

Deliverable name	Deadline
Management Plan with annexed Monitoring protocol	10/2016
After LIFE Plan	03/2020
Project's Final technical report	03/2020
External Audit	03/2020

E1's PROJECT MILESTONES

Milestone name	Deadline
Kick off Meeting and Presentation of the management Staff	09/2016
Final meeting of LIFE Agrowetlands II	02/2020

DELIVERABLE PRODUCTS OF THE PROJECT

Name of the Deliverable	Number of the associated action	Deadline
Management Plan with annexed Monitoring protocol	E 1	28/10/2016
Agrowetland Communication Plan	D 1	30/11/2016
Press release n°1 Agrowetlands Kick-Off	D 1	30/11/2016
Project Website	D 1	30/11/2016
Geo-database with the main soil, climate, watertable, water salinity, water flow parameters	B 1	30/06/2017
Information Board	D 1	15/09/2017
New model code for water management linked to the geo-database	B 2	15/09/2017
Project's Brochures and Posters	D 1	15/09/2017
updated LIFE Project Specific Indicators Call excel table	C 1	22/12/2017
User Manual and description of model functionality	B 2	15/02/2018
Installation and User Manual of the Agrowetland Unit	B 4	27/04/2018
Report on the integrated Agrowetland WSN, including functionality tests	B 4	27/04/2018
updated LIFE Project Specific Indicators Call excel table	C 1	15/06/2018
Press release n° 2 - on the Start of the AGROWETLAND system operation	D 1	16/07/2018
Report on AGROWETLANDS water management model	B 2	31/08/2018
Report on the Agrowetlands web-platform	B 3	31/08/2018
Web-based User interface user manual	B 3	31/08/2018
Short report about the correct operation of wireless sensors network	B 5	07/09/2018

Scientific Paper published (agronomic aspect project implementation)	D 2	21/12/2018
Scientific Paper published (engineering aspect project implementation)	D 2	21/12/2018
updated LIFE Project Specific Indicators Call excel table	C 1	30/06/2019
Report with the main results f the socioeconomic poll to farmers	C 2	20/12/2019
Scientific Paper published (agronomic aspect project results)	D 2	27/12/2019
Scientific Paper published (engineering aspect project results)	D 2	27/12/2019
Press release n° 3 - Agrowetlands Final Conference announcement and results	D 1	31/01/2020
SMART AGROWETLANDS II system report: closed loop behavior, assessment of performance	B 6	31/01/2020
Complete life-cycle assessment on positive and negative environmental impacts of LIFE AGROWETLANDS II system for water and soil salinity management	C 1	28/02/2020
Final Report of evaluation of water use efficiency, soil salinity and salts load to the wetland within the pilot area	C 1	28/02/2020
Report on transferability of the AGROWETLANDS II system in the Comunidad de Carrizales	D 2	28/02/2020
Report: Assessment on socio-economic impact	C 2	28/02/2020
After LIFE Plan	E 1	27/03/2020
External Audit	E 1	27/03/2020
Guide for recommended measures aimed at the improvement of water quality and quantity of freshwater ponds in similar agro-wetlands	C 2	27/03/2020
Project's Final technical report	E 1	27/03/2020
Report on the improvement of the water quality parameters along the development of the project and its relationship with the irrigation scheduling followed in the Agrisfera's district	C 2	27/03/2020

Layman's report	D 1	31/03/2020
final LIFE Project Specific Indicators Call excel table	C 1	30/09/2020

MILESTONES OF THE PROJECT

Name of the Milestone	Number of the associated action	Deadline
Kick off Meeting and Presentation of the management Staff	E 1	23/09/2016
Project's website online	D 1	30/11/2016
First set of sensors ready for the installation at action B6	B 4	10/03/2017 the beginning of
First set of nodes and gateways for the WSN	B 5	31/05/2017 preliminary functioning of the
Initial spatial evaluation of soil salinity, water crops, watr flow and salinity in irrigation and drainage channels within each sector of the pilot area	B 1	01/07/2017 table level,
model of soil and water salinity for irrigation version)	B 2	31/07/2017 recommendation (first
Spatial delimitation of sectors for water	B 1	15/09/2017 management
Preliminary version of web control system	B 3	29/09/2017 available
Definition of spatial localization of sensors within	B 1	16/10/2017 the sectors
model of soil and water salinity for irrigation recommendation (second version) checked at small scale in crops cultivated in pots	B 2	15/01/2018
Fully integrated heterogeneous WSN is ready for installation in the Agrisfera agro-wetland	B 4	27/04/2018 the
Agrowetland web- platform (web-control system interface for irrigation management) final version	B 3	16/07/2018 and user up and running
Start of the AGROWETLAND system operation	B 6	16/07/2018
model of soil and water salinity for irrigation recommendation (final version) checked at field scale in pilot area	B 2	27/07/2018
Wireless sensors network and weather station	B 5	31/07/2018 fully installed
Full AGROWETLANDS system operating in the adaptation and validation)	B 6	16/09/2019 pilot area (after
Final Conference of the Project	D 1	07/02/2020
Final meeting of LIFE Agrowetlands II	E 1	21/02/2020

ACTIVITY REPORTS FORESEEN

Please indicate the deadlines for the following reports:

- Progress Reports n°1, n°2 etc. (if any; to ensure that the delay between consecutive reports does not exceed 18 months)

- Mid term report payment request (for project longer than 24 months or with Eu contibution of more than EUR300,000)
- Final Report with payment request (to be delivered within 3 months after the end of the project)

Type of report	Deadline
Progress report	22/12/2017
Midterm report	15/06/2018
Progress report	30/06/2019
Final report	30/09/2020

TIMETABLE

Action		2016				2017				2018				2019				2020				2021			
Action numbe	Name of the action	I	II	III	IV	I	II	III	IV	I	II	III	IV	I	II	III	IV	I	II	III	IV	I	II	III	IV
A. Preparatory actions (if needed)																									
B. Implementation actions (obligatory)																									
B.1	Environmental database construction, modelling of water and salt flow in the pilot area			■	■	■	■	■																	
B.2	Development of a new model of soil and water salinity for irrigation recommendation				■	■	■	■	■	■	■	■													
B.3	Development and implementation of web-control system and user interface for irrigation management				■	■	■	■	■	■	■	■													
B.4	Development of an integrated WSN modules to monitor soil and water salinity, and water-table level for the sustainable irrigation				■	■	■	■	■	■	■														
B.5	Field installation of wireless sensor network in pilot area							■	■	■	■	■													
B.6	Field application of SMART-AGROWETLANDS II and evaluation of its effects on crops						■	■	■	■	■	■	■	■	■	■	■	■	■						
C. Monitoring of the impact of the project actions (obligatory)																									
C.1	Monitoring the Project’s impact on salinity status in agro-wetland pilot area								■	■	■	■	■	■	■	■	■	■	■						
C.2	Monitoring the Project’s impact on the natural and semi-natural habitats, plant communities and biological quality of water and assessment on socio-economic impact					■	■	■	■	■	■	■	■	■	■	■	■	■							
D. Public awareness and dissemination of results (obligatory)																									
D.1	Communication Pack and Dissemination Activities			■	■	■	■	■	■	■	■	■	■	■	■	■	■	■							
D.2	Networking, Training activities and Dissemination Events - Replicability			■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■						
E. Project management (obligatory)																									
E.1	Project management, monitoring and After Life Plan			■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■						